ForeverMart

Software Quality Assurance Report

Software Testing - Portfolio Part 3

Submitted by

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Course

Software Quality Engineering and Testing

Project under test

https://forevermart.vercel.app/

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Table of contents

Section	Page
Complete bug register	3-8
Observation register	9-10
Executive summary	11
Scope and test environment	12
Test approach and acceptance criteria	13
How to read the report	14
Detailed bug and observation records	15-104
Requirement and defect traceability	105
Checkout decision table	106
Transaction workflow and control boundary	107
Defect severity and priority	108
Pareto analysis of supported defects	109
Defect distribution by application area	110
Status and measurement boundaries	111
Cause and effect investigation	112
Defect lifecycle and triage	113
Continuous improvement plan	114
Corrective action and regression matrix	115
Screenshot evidence	116-124
API and persistence evidence	125
Reference book crosswalk	126
Handover and supporting references	127
Techniques, tools and reference sources	128

Complete bug register

ID	Bug / finding	Severity	Details
FM-001	Shop Now does not open the catalogue	Medium	15
FM-002	Best Seller section has no products despite flagged catalogue items	Medium	16
FM-004	Reviews tab does not display reviews	Medium	18
FM-005	Product rating and review count are hard coded	Medium	19
FM-006	Jacket description describes a short sleeved T-shirt	Medium	20
FM-007	Detailed Description panel explains ecommerce instead of the product	Low	21
FM-008	Product image and title describe different garment styles	Medium	22
FM-009	Related Products includes the product already being viewed	Low	23
FM-010	Selected size carries over when navigating to a related product	Medium	24
FM-011	Unknown product leaves an incomplete purchase page	Medium	25
FM-012	Search with no matches displays an unexplained empty grid	Medium	26
FM-013	Leading and trailing spaces prevent otherwise valid search matches	Medium	27

Complete bug register - continued

ID	Bug / finding	Severity	Details
FM-014	Collection filters and sorting are lost after returning from a product	Medium	28
FM-015	Footer navigation labels cannot be opened	Medium	29
FM-016	Forgot your password has no recovery action	Medium	30
FM-017	My Profile menu item has no destination	Medium	31
FM-018	Explore Jobs does not open career information	Low	32
FM-019	Newsletter subscription produces no result	Medium	33
FM-020	Product names are absent from cart and order rows	Medium	34
FM-021	Guest cart is overwritten after account creation	High	35
FM-022	An empty cart can be submitted as an order	High	36
FM-023	Empty-cart total does not agree with displayed shipping	Medium	37
FM-024	Whitespace-only delivery information is accepted	High	38
FM-025	Negative phone number is accepted at checkout	High	39

Complete bug register - continued

ID	Bug / finding	Severity	Details
FM-026	Negative postal code is accepted at checkout	Medium	40
FM-027	Order endpoint accepts a malformed delivery email	Medium	41
FM-028	Order endpoint accepts a negative fractional quantity	High	42
FM-029	Order endpoint accepts a size not sold for the product	High	43
FM-030	Order endpoint accepts a non-existent product ID	High	44
FM-031	Order endpoint trusts an altered client line-item price	High	45
FM-032	Order endpoint trusts the submitted total amount	High	46
FM-033	Cart endpoint stores a negative fractional quantity	High	47
FM-034	Cart endpoint accepts an unsupported size	High	48
FM-035	Cart endpoint accepts a non-existent product	High	49
FM-036	Guest checkout loses the address when authentication is required	Medium	50
FM-037	Postal field strips letters from an alphanumeric code	Medium	51

Complete bug register - continued

ID	Bug / finding	Severity	Details
FM-038	Phone field rejects a normally formatted telephone string	Medium	52
FM-039	Unknown route has no not-found message or recovery	Medium	53
FM-040	Profile response includes password and refresh-token fields	High	54
FM-041	Search opener cannot be activated with the keyboard	Medium	55
FM-042	Search close icon lacks a keyboard action and name	Medium	56
FM-043	Authenticated account menu is not keyboard operable	Medium	57
FM-044	Login and Create Account mode switch is not a keyboard control	Medium	58
FM-045	Product gallery thumbnails are pointer-only controls	Medium	59
FM-046	Remove cart item is an unnamed pointer-only image	Medium	60
FM-047	Cart product navigation is available only through a clickable image	Medium	61
FM-048	Mobile menu opener has no button semantics	Medium	62
FM-049	Mobile menu Back affordance is not keyboard operable	Medium	63

Complete bug register - continued

ID	Bug / finding	Severity	Details
FM-050	Closed mobile menu leaves offscreen links in the accessibility tree	Medium	64
FM-051	Mobile menu does not manage focus or dismiss on Escape	Medium	65
FM-052	Mobile filter disclosure is a paragraph rather than a control	Medium	66
FM-053	Payment methods are not keyboard selectable radio controls	Medium	67
FM-054	Stripe option has no accessible payment-provider name	Medium	68
FM-055	Selected product size is communicated only by colour	Medium	69
FM-056	Collection sort selector has no associated label	Medium	70
FM-057	Delivery fields rely on disappearing placeholders	Medium	71
FM-058	Authentication fields have no persistent associated labels	Medium	72
FM-059	Personal-data inputs do not expose their autocomplete purposes	Medium	73
FM-060	Muted home-page text has insufficient contrast	Medium	74
FM-061	Newsletter input suppresses its visible keyboard focus	Medium	75

Complete bug register - continued

ID	Bug / finding	Severity	Details
FM-062	Meaningful product photographs have empty alternative text	Medium	76
FM-063	Visible page and section titles are not semantic headings	Medium	77
FM-064	Page content lacks main and footer landmarks	Low	78
FM-065	Browser title does not identify the current page	Low	79
FM-066	Layout overflows horizontally at 320 CSS pixels	Medium	80
FM-067	Home-page hero misspells Arrivals	Low	81
FM-068	Template filler text remains in customer-facing sections	Low	82
FM-069	Contact details conflict between the footer and Contact page	Medium	83
FM-070	About page contains an incomplete support sentence	Low	84
FM-079	Newsletter placeholder contains incorrect wording	Low	93

Observation register

ID	Bug / finding	Severity	Details
FM-003	Latest Collection is not ordered by creation date	Low	17
FM-071	Winterwear filter has no matching catalogue classification	Low	85
FM-072	Empty cart has no explanatory state or shopping recovery	Low	86
FM-073	Checkout does not show the products being purchased	Medium	87
FM-074	Order history omits order identifiers	Medium	88
FM-075	Order history loses order-level grouping	Medium	89
FM-076	Order history omits the final order total and shipping breakdown	Medium	90
FM-077	Order history has no delivery-address review	Low	91
FM-078	About page advertises product ranges absent from the inspected catalogue	Low	92
FM-080	Copyright footer contains incomplete grammar and inconsistent branding	Low	94
FM-081	Public document has no Content Security Policy	Medium	95
FM-082	Public document lacks an explicit framing restriction	Medium	96

Observation register - continued

ID	Bug / finding	Severity	Details
FM-083	Public responses omit the nosniff hardening header	Low	97
FM-084	Public document has no explicit Referrer Policy	Low	98
FM-085	Offscreen catalogue images are not marked for lazy loading	Low	99
FM-086	Product images do not advertise responsive source variants	Low	100
FM-087	Public pages lack a page-specific meta description	Low	101
FM-088	Footer email and telephone cannot be activated directly	Low	102
FM-089	Checkout does not state which dollar currency is used	Low	103
FM-090	Displayed return and exchange promises lack instructions	Low	104

Executive summary

ForeverMart supports basic catalogue browsing, variant selection, cart use and cash-on-delivery ordering. The audit found material weaknesses in server validation and pricing, incomplete storefront actions, incorrect product content, and accessibility barriers.

Result	Count	Meaning
Supported defects	70	Reproduced behaviour or directly inspected implementation conflicts with the stated audit criterion.
Improvement observations	20	Observed gaps needing product-owner review; excluded from the defect total.
Requested defect target	100	30 additional distinct defects were not established by this audit.
Fixed or closed defects	0	No application repairs were made or retested.

Most urgent corrective work

The order API accepted altered prices and totals, a fabricated product ID, an unsupported size, a negative fractional quantity and a malformed delivery email. The UI also accepted an empty order and unusable address data. These behaviours were confirmed using the authorised disposable account and COD test orders.

Readiness opinion

Recommend a hold on real customer ordering until the High-severity records are corrected and independently retested. This recommendation is about the tested build and does not claim that every part of the application is broken.

Evidence and limits

Evidence includes page captures, DOM snapshots, public product data, a publicly served client bundle, HTTP headers and responses for the audit account only. No administrator functions, real card payments, load tests, other-user records or server source code were tested.

Report structure adapts Lewis Appendix E29, pp. 384-385 (PDF pp. 411-412), and the E11 analysis and readiness sections, pp. 361-362 (PDF pp. 388-389).

Scope and test environment

Area	Audit scope
Application	Public ForeverMart storefront and its observed application API
Build identity	Client asset index-r3xEth_L.js; the evidence manifest records its SHA-256. No release number or commit was provided.
Browser	Codex in-app browser; desktop, 390 x 844 and 320 x 740 CSS-pixel viewport checks
Test date	9 October 2026; saved API/header evidence uses UTC timestamps
Account	One disposable demo account; synthetic example.com contact data
Transactions	Eight authorised COD test orders; no real payment
Methods	Exploratory browser testing, boundary/negative API checks, DOM accessibility inspection and public client-code review

Included

Home, Collection, search, filters, sort, product details, gallery and sizes, cart, Signup/Login, checkout, own order history, About, Contact, shared navigation, narrow-screen layout and selected response-hardening settings.

Excluded

Admin operations, payment settlement and Stripe purchase completion, refunds, stock deduction, email delivery, rate-limit/load testing, cross-browser certification, full screen-reader certification, database access and source-level unit coverage.

Authorisation and cleanup

The owner confirmed this is a demo site and permitted disposable accounts and test orders. Tests accessed only the audit account. The two fabricated invalid cart entries were removed. The account and eight demo orders remain as test evidence; no other-user data was changed.

Adapted from Lewis E4 client/server and Internet test plan, pp. 353-354 (PDF pp. 380-381), and E21 test strategy, pp. 374-375 (PDF pp. 401-402).

Test approach and acceptance criteria

The course brief specifies testing, defect documentation and remedies, but supplies no detailed ForeverMart requirements. The audit therefore uses explicit ecommerce correctness rules, the behaviour advertised by the interface, and selected accessibility criteria as its baseline. Product choices and hardening recommendations are labelled observations where acceptance requires owner review.

Technique	Representative data or check	Purpose
Normal flow	Existing product, size M, quantity 1; browse, sort and cart	Establish usable baseline behaviour
Boundary and negative values	Empty items, three spaces, -1 contact fields, quantity -0.5	Challenge required fields and integer/domain constraints
Identity and variant checks	Unknown product ID; QA-NOT-A-SIZE	Check references against the server catalogue
Price integrity	$200 item with price changed to 1 or total changed to 1	Check server ownership of monetary calculations
State transitions	Guest to Signup; related product; browser Back	Check selection and draft persistence
Accessibility and responsive checks	Keyboard semantics, labels, contrast, 320 CSS px reflow	Check operability and meaningful presentation

Severity and priority

High: monetary or data-integrity risk, lost purchase state, or unnecessary authentication-data exposure. Medium: important function, content or accessibility failure. Low: limited content, structural or polish impact. P1, P2 and P3 are proposed fix order corresponding to these levels; they are not agreed delivery dates.

Entry and exit conditions

Entry was met when the public deployment and demo test permission were available. Proposed release exit: no unresolved High defects; all remediated journeys pass positive and negative regression; remaining risks are explicitly accepted by the project owner.

Techniques adapt Lewis G4 Boundary Value Testing, p. 453 (PDF p. 480), G13 Equivalence Partitioning, p. 493 (PDF p. 520), and G22 Positive and Negative Testing, p. 501 (PDF p. 528).

How to read the report

Detailed records

Each record identifies the finding, test reference, affected module, URL, method, severity, priority, preconditions, reproduction steps, expected and actual result, impact, remedy, retest acceptance and evidence.

Defects and observations

The bug register contains 70 confirmed defects. The observation register contains 20 additional quality gaps needing an owner decision. The 90 detailed records preserve their original FM identifiers; observations are excluded from the confirmed defect count.

Status and ownership

Open means a defect is reported without a verified repair. Review required means an observation needs a product decision. Proposed owner roles have not been assigned to named people. Developer and client comments remain Not supplied.

Classification discipline

Repeated symptoms of a shared implementation are grouped where appropriate. Separate server rules for quantity, product identity, variants, price and total retain separate records because each needs its own validation and regression assertion.

Evidence and corrective action

Evidence paths refer to the original captures and sanitised responses in the audit folder. Recommended remedies and retest criteria describe the next corrective work; no record is claimed to be fixed or closed.

FM-001 | Shop Now does not open the catalogue

DEFECT | Functionality

Module	Home	Severity / priority	Medium / P2
Test reference	TC-001	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the home page.
2. Click SHOP NOW in the hero.

Expected result	Actual result
The call to action opens the collection or another defined shopping destination.	The URL and page remain unchanged; SHOP NOW is rendered as a paragraph without an action.

Impact The primary shopping invitation cannot be used.

Recommended remedy

Render a real link to /collection with a visible focus state.

Retest acceptance

Mouse click and Enter on the focused link both open Collection.

Evidence

evidence/shop-now-no-action.json; evidence/shop-now-no-action.jpg; raw/app.js

FM-002 | Best Seller section has no products despite flagged catalogue items

DEFECT | Functionality

Module	Home	Severity / priority	Medium / P2
Test reference	TC-002	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Browser test and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Allow the catalogue to finish loading.
2. Inspect BEST SELLER and compare the public product flags.

Expected result	Actual result
Products marked bestSeller are displayed in the section.	The section is empty although five catalogue records have bestSeller=true. The public bundle filters the different key bestseller.

Impact Merchandising content disappears and an empty section occupies the home page.

Recommended remedy

Use the product contract key bestSeller and map productName consistently. Add a populated-fixture component check.

Retest acceptance

The five flagged records render with names and prices; an empty source has a deliberate empty state.

Evidence

evidence/home-desktop.jpg; raw/products.json; raw/app.js

FM-003 | Latest Collection is not ordered by creation date

OBSERVATION | Content

Module	Home	Severity / priority	Low / P3
Test reference	TC-003	Status	Review required
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/

Method Public data and code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect the ten Latest Collection items.
2. Compare their createdAt values with the full public catalogue.

Expected result	Actual result
A section labelled Latest Collection presents the newest products first, or explains its curation.	The list takes the first ten API records. An October 14 record leads the list while newer October 22 records are available.

Impact The section title does not describe its actual selection.

Recommended remedy

Sort a copy of products by createdAt descending before taking ten, or rename the section Featured Collection.

Retest acceptance

A fixture containing older and newer products displays the declared ordering.

Evidence

raw/products.json; raw/app.js; evidence/home-desktop.jpg

Audit note Latest is an inferred merchandising criterion; confirm with the owner.

FM-004 | Reviews tab does not display reviews

DEFECT | Functionality

Module	Product	Severity / priority	Medium / P2
Test reference	TC-004	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the product and allow its data to load.
2. Click Reviews (122).

Expected result	Actual result
The selected tab displays the reviews or an honest unavailable state.	The displayed content does not change; the tab is a paragraph without a click handler.

Impact Customers cannot inspect the reviews advertised by the page.

Recommended remedy

Implement accessible tabs backed by review data, or remove the inactive review tab and count.

Retest acceptance

Selecting Reviews changes the panel; keyboard selection works and the empty state is truthful.

Evidence

evidence/reviews-no-change.jpg; raw/app.js

FM-005 | Product rating and review count are hard coded

DEFECT | Content

Module	Product	Severity / priority	Medium / P2
Test reference	TC-005	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method Browser test and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the cotton T-shirt.
2. Open the Women Zip-Front Jacket and compare the rating area with the public data and bundle.

Expected result	Actual result
Rating and count derive from actual product reviews, or are clearly identified as sample content.	Both display the same stars and (122); these values are fixed in the public component and not derived from review data.

Impact The storefront presents unsupported customer-feedback claims.

Recommended remedy

Aggregate approved reviews by product. Hide ratings when no verified review source exists; label demo fixtures explicitly.

Retest acceptance

Products with zero, one and several reviews show the correct counts and calculated ratings.

Evidence

evidence/product-details.json; evidence/second-product-loaded.json; raw/app.js

FM-006 | Jacket description describes a short sleeved T-shirt

DEFECT | Content

Module	Product	Severity / priority	Medium / P2
Test reference	TC-006	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/product/68f38f15f780484273055b41

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open Women Zip-Front Relaxed Fit Jacket priced $399.
2. Read the short product description.

Expected result	Actual result
The description explains the jacket actually offered.	The jacket uses a generic round-neck, short-sleeved pullover shirt description. The same text is reused across different apparel types.

Impact Customers receive incorrect information about the item.

Recommended remedy

Replace copied descriptions with product-specific material, fit, closure and care details; validate catalogue content before publishing.

Retest acceptance

Jackets, trousers and tops each have accurate descriptions matching their records and images.

Evidence

evidence/second-product-loaded.json; evidence/second-product-loaded.jpg; raw/products.json

FM-007 | Detailed Description panel explains ecommerce instead of the product

DEFECT | Content

Module	Product	Severity / priority	Low / P3
Test reference	TC-007	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the product.
2. Read the Description panel below the purchase controls.

Expected result	Actual result
The panel contains useful details about this product.	The text explains what an ecommerce website is rather than describing the selected apparel.

Impact A prominent information panel gives customers no product details.

Recommended remedy

Populate the panel from a product detail field; remove template instructional copy.

Retest acceptance

Changing product changes the details and no ecommerce-definition template text remains.

Evidence

evidence/product-details.json; raw/app.js

FM-008 | Product image and title describe different garment styles

DEFECT | Content

Module	Product	Severity / priority	Medium / P2
Test reference	TC-008	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open Men Round Neck Pure Cotton T-shirt priced $200.
2. Compare the main photo with the title and description.

Expected result	Actual result
The images depict the named round-neck, short-sleeved T-shirt.	The main photo depicts a pink collared, long-sleeved shirt while the title and description describe a round-neck T-shirt.

Impact The shopper cannot reliably determine which garment will be delivered.

Recommended remedy

Correct the product-to-image association or the product text after checking the intended inventory item.

Retest acceptance

All gallery images match the approved product identity and garment characteristics.

Evidence

evidence/product-details.jpg; evidence/product-details.json; raw/products.json

FM-009 | Related Products includes the product already being viewed

DEFECT | Functionality

Module	Product	Severity / priority	Low / P3
Test reference	TC-009	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method Browser test and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the $200 T-shirt.
2. Inspect Related Products.

Expected result	Actual result
Recommendations provide other relevant items.	The current product appears in its own related-product list; the filter does not exclude the current ID.

Impact A recommendation slot is wasted on an item already open.

Recommended remedy

Exclude the current product ID before limiting related results.

Retest acceptance

No related result has the current ID, including after navigation between related items.

Evidence

evidence/product-details.json; raw/app.js

FM-010 | Selected size carries over when navigating to a related product

DEFECT | Functionality

Module	Product	Severity / priority	Medium / P2
Test reference	TC-010	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Choose size M on the $200 T-shirt.
2. Open a related product priced $280.
3. Inspect the size selection before choosing a size again.

Expected result	Actual result
The new product starts with no selection, or explicitly asks the customer to confirm its variant.	M remains selected on the new product even though it was selected for the previous item.

Impact A shopper can add the next item using an unintended variant.

Recommended remedy

Reset variant and gallery state whenever productId changes; validate the selected size against the new product.

Retest acceptance

Navigate between products with different size lists; adding requires a valid selection for the new item.

Evidence

evidence/related-size-carryover.json; evidence/related-size-carryover.jpg

FM-011 | Unknown product leaves an incomplete purchase page

DEFECT | Functionality

Module	Product	Severity / priority	Medium / P2
Test reference	TC-011	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/product/000000000000000000000000

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the stated non-existent product ID.
2. Allow the request to complete.

Expected result	Actual result
A Product not found state replaces purchase controls and offers a route back to the collection.	The page shows an empty title, a currency symbol without a price, hard-coded review content and ADD TO CART.

Impact A broken URL looks like a purchasable item rather than a handled missing record.

Recommended remedy

Track loading, found, missing and failed states; render purchase controls only for a valid product.

Retest acceptance

Missing and malformed IDs have a clear recovery page; valid products still load normally.

Evidence

evidence/missing-product.json; evidence/missing-product.jpg

FM-012 | Search with no matches displays an unexplained empty grid

DEFECT | Usability

Module	Collection	Severity / priority	Medium / P2
Test reference	TC-012	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open search on Collection.
2. Enter zzqa-no-such-product.

Expected result	Actual result
A no-results message identifies the search and offers a clear or reset action.	No product cards appear and no no-results explanation is displayed.

Impact Users cannot distinguish an empty result from a failed load.

Recommended remedy

Render a no-results state after loading completes, with a clear-search action.

Retest acceptance

An impossible query shows the message; clearing it restores products.

Evidence

evidence/search-no-results.json; evidence/search-no-results.jpg

FM-013 | Leading and trailing spaces prevent otherwise valid search matches

DEFECT | Functionality

Module	Collection	Severity / priority	Medium / P2
Test reference	TC-013	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method Browser test and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Search for cotton.
2. Replace it with two leading spaces, cotton, and two trailing spaces.

Expected result	Actual result
Incidental outside whitespace does not change the meaningful query.	cotton finds 22 items, while the padded query finds none. The filter lowercases text but does not trim it.

Impact Pasted or casually typed queries produce false empty results.

Recommended remedy

Normalize the search with trim and consistent case handling before filtering.

Retest acceptance

cotton, Cotton and padded cotton return the same matching IDs.

Evidence

evidence/search-spaces.json; evidence/search-spaces.jpg; raw/app.js

FM-014 | Collection filters and sorting are lost after returning from a product

DEFECT | Functionality

Module	Collection	Severity / priority	Medium / P2
Test reference	TC-014	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Select Men and Sort by Low to High.
2. Open a product.
3. Use browser Back.

Expected result	Actual result
The previous collection criteria are restored.	Men is unchecked and sorting returns to Relevant; all products are shown.

Impact Customers must repeat their selection after every product inspection.

Recommended remedy

Store search, category and sort criteria in URL parameters or durable route state.

Retest acceptance

Back restores the same criteria and results, and a shared filtered URL opens the same view.

Evidence

evidence/collection-criteria.json; evidence/collection-after-back.json

FM-015 | Footer navigation labels cannot be opened

DEFECT | Functionality

Module	Shared layout	Severity / priority	Medium / P2
Test reference	TC-015	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Go to the footer.
2. Attempt Home, About Us, Delivery and Privacy Policy.

Expected result	Actual result
Each advertised navigation destination is an operable link.	All four labels are static list items with no href or action.

Impact Customers cannot reach the pages promised by the footer.

Recommended remedy

Create valid links and publish the intended delivery and privacy content. Grouped here because the same footer implementation affects all four labels.

Retest acceptance

All four destinations work with mouse and keyboard and have meaningful URLs.

Evidence

evidence/home-mobile.json; raw/app.js

FM-016 | Forgot your password has no recovery action

DEFECT | Functionality

Module	Account	Severity / priority	Medium / P2
Test reference	TC-016	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/login

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the account form.
2. Click Forgot your password?

Expected result	Actual result
A recovery flow or a clearly explained support alternative opens.	The page stays unchanged; the text has a pointer cursor but no handler.

Impact A user who has forgotten a password has no advertised recovery path.

Recommended remedy

Implement a verified recovery process, or replace the inactive control with an accurate support instruction.

Retest acceptance

The control opens recovery; account existence is not disclosed by recovery responses.

Evidence

evidence/forgot-password-no-action.json; evidence/forgot-password-no-action.jpg; raw/app.js

FM-017 | My Profile menu item has no destination

DEFECT | Functionality

Module	Account	Severity / priority	Medium / P2
Test reference	TC-017	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Sign in with the disposable account.
2. Open the profile menu and click My Profile.

Expected result	Actual result
The profile page opens or the unavailable feature is not presented as an action.	No page or panel opens; My Profile is static text.

Impact The account menu advertises a function users cannot use.

Recommended remedy

Link to an implemented profile page or remove the inactive menu item until available.

Retest acceptance

Selecting My Profile opens a working destination with the authenticated demo profile.

Evidence

evidence/profile-no-action.json; evidence/profile-no-action.jpg

FM-018 | Explore Jobs does not open career information

DEFECT | Functionality

Module	Contact	Severity / priority	Low / P3
Test reference	TC-018	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/contact

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open Contact.
2. Click Explore Jobs.

Expected result	Actual result
The button opens career details or a supported application destination.	The page and URL remain unchanged.

Impact The careers call to action is nonfunctional.

Recommended remedy

Provide an actual careers link, or remove the button when no vacancies or careers page exist.

Retest acceptance

Mouse and keyboard activation open the declared destination.

Evidence

evidence/jobs-no-action.json; evidence/jobs-no-action.jpg; raw/app.js

FM-019 | Newsletter subscription produces no result

DEFECT | Functionality

Module	Home	Severity / priority	Medium / P2
Test reference	TC-019	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Browser test and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Enter a syntactically valid synthetic email in Subscribe now.
2. Click SUBSCRIBE.

Expected result	Actual result
The form records the request and shows success or a useful error.	There is no feedback or subscription action. The handler only prevents the default form submission.

Impact The advertised 20 percent offer has no usable subscription path.

Recommended remedy

Connect the form to the intended subscription service and show a result; remove the offer until its fulfilment works.

Retest acceptance

Valid, duplicate and invalid requests produce truthful, accessible feedback without unintended subscriptions.

Evidence

evidence/subscribe-no-feedback.json; evidence/subscribe-no-feedback.jpg; raw/app.js

FM-020 | Product names are absent from cart and order rows

DEFECT | Functionality

Module	Cart	Severity / priority	Medium / P2
Test reference	TC-020	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/cart

Method Browser test and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Add the $200 T-shirt in size M.
2. Inspect its cart row, then the own demo order history.

Expected result	Actual result
Each row identifies the product by its stored productName.	Images, price and size appear but the product name is blank. These views read name while records contain productName.

Impact Customers cannot clearly identify line items; the same data-contract mismatch affects both views.

Recommended remedy

Use one typed product contract and read productName in cart and order components.

Retest acceptance

Names appear for all line items in both cart and order history.

Evidence

evidence/cart-one-item.json; evidence/orders-validation-tests.json; raw/app.js

FM-021 | Guest cart is overwritten after account creation

DEFECT | Functionality

Module	Cart	Severity / priority	High / P1
Test reference	TC-021	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/cart

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. As a guest, add one item.
2. Create the disposable account.
3. Inspect the cart after authentication.

Expected result	Actual result
The guest selection survives or is merged with the account cart.	The badge changes from one item to zero and the guest item disappears when the empty account cart is loaded.

Impact A customer loses the prepared purchase at the authentication step.

Recommended remedy

Merge validated guest items with the account cart on authentication and reconcile the result before replacing local state.

Retest acceptance

Guest-only and existing-account carts merge without loss or duplicate quantities.

Evidence

evidence/cart-one-item.json; evidence/signup-outcome.json; raw/app.js

FM-022 | An empty cart can be submitted as an order

DEFECT | Data integrity

Module	Checkout	Severity / priority	High / P1
Test reference	TC-022	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Use the authenticated demo account with an empty cart.
2. Proceed to checkout, complete the delivery form and submit cash on delivery.

Expected result	Actual result
Checkout is blocked when there are no valid line items; the server rejects an empty items array.	Order creation succeeds and the own-orders response contains an order with items=[] and amount=10.

Impact Invalid transactions are stored and can enter fulfilment workflows. Empty orders are also invisible in the flattened history.

Recommended remedy

Enforce non-empty validated items on the server and block the empty-cart checkout path in the UI.

Retest acceptance

An empty order returns a validation error and creates no record; one valid item succeeds.

Evidence

evidence/cart-empty.json; evidence/empty-order-outcome.json; raw/demo-orders.json

FM-023 | Empty-cart total does not agree with displayed shipping

DEFECT | Functionality

Module	Cart	Severity / priority	Medium / P2
Test reference	TC-023	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/cart

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open an empty cart.
2. Compare Subtotal, Shipping Fee and Total.

Expected result	Actual result
Displayed amounts reconcile, and no charge applies to an order that cannot be placed.	Subtotal is $0.00 and shipping $10.00, but Total is $0.00. An empty submitted order instead stores $10.

Impact The stated payable amount is internally inconsistent.

Recommended remedy

Use one totals function across display and submission; hide or zero shipping for an empty cart.

Retest acceptance

Empty and populated carts reconcile; submission and receipt equal the final displayed total.

Evidence

evidence/cart-empty.json; evidence/cart-empty.jpg; raw/demo-orders.json

FM-024 | Whitespace-only delivery information is accepted

DEFECT | Data integrity

Module	Checkout	Severity / priority	High / P1
Test reference	TC-024	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Enter three spaces in first name, last name, street, city, state and country; use a valid synthetic email.

2. Submit a demo COD order.

Expected result	Actual result
Required address strings remain non-empty after trimming and are rejected otherwise.	The order is accepted; the persisted address contains whitespace-only values.

Impact The stored order has no deliverable recipient or address.

Recommended remedy

Trim and validate required strings on both client and server; show errors beside the fields.

Retest acceptance

Each whitespace-only required field rejects the order without persistence; a valid address succeeds.

Evidence

evidence/checkout-invalid-values.jpg; raw/demo-orders.json

FM-025 | Negative phone number is accepted at checkout

DEFECT | Data integrity

Module	Checkout	Severity / priority	High / P1
Test reference	TC-025	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Complete other delivery fields and enter -1 for Phone number.
2. Submit the authorized demo COD test order.

Expected result	Actual result
The contact field accepts a valid telephone string and rejects an impossible negative number.	The server stores phoneNumber="-1".

Impact The fulfilment team has no usable delivery contact.

Recommended remedy

Use type=tel and server-side phone validation appropriate to the supported countries; do not treat phone numbers as arithmetic quantities.

Retest acceptance

-1 and malformed phones are rejected; supported formatted numbers remain intact.

Evidence

evidence/checkout-invalid-values.jpg; raw/demo-orders.json

FM-026 | Negative postal code is accepted at checkout

DEFECT | Data integrity

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-026	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Complete the other delivery fields and enter -1 for Zipcode.
2. Submit the authorized demo COD order.

Expected result	Actual result
Postal codes are validated as country-appropriate identifiers.	The persisted order contains zipCode="-1".

Impact The order can contain unusable delivery routing data.

Recommended remedy

Use a text postal-code field and validate it against the selected supported country on the server.

Retest acceptance

Negative values fail; valid supported numeric and alphanumeric codes pass.

Evidence

evidence/checkout-invalid-values.jpg; raw/demo-orders.json

FM-027 | Order endpoint accepts a malformed delivery email

DEFECT | Data integrity

Module	Order API	Severity / priority	Medium / P2
Test reference	TC-027	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/order/placeOrder

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Authenticate as the disposable account.
2. Submit one valid COD line item with address.email=not-an-email.
3. Fetch own orders.

Expected result	Actual result
The server rejects a malformed email independently of browser validation.	HTTP 200 success is returned and the malformed email persists in the order.

Impact API clients can create orders with unusable notification addresses.

Recommended remedy

Apply the same validated address schema at the API boundary; return a field-specific 4xx error.

Retest acceptance

Malformed email creates no order; a valid synthetic email passes.

Evidence

raw/api-validation.json

FM-028 | Order endpoint accepts a negative fractional quantity

DEFECT | Data integrity

Module	Order API	Severity / priority	High / P1
Test reference	TC-028	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/order/placeOrder

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Use the demo account and a valid product ID.
2. Submit quantity=-0.5 in a COD order.
3. Read back own orders.

Expected result	Actual result
Every apparel quantity is a positive integer within the allowed limit.	The endpoint returns success and persists -0.5; the history displays that quantity.

Impact Invalid units corrupt order and fulfilment calculations.

Recommended remedy

Validate Number.isSafeInteger(quantity), quantity>=1 and a documented maximum before any write.

Retest acceptance

-0.5, 0, fractional and excessive values fail; 1 and a valid upper bound pass.

Evidence

raw/api-validation.json; evidence/orders-validation-tests.json

FM-029 | Order endpoint accepts a size not sold for the product

DEFECT | Data integrity

Module	Order API	Severity / priority	High / P1
Test reference	TC-029	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/order/placeOrder

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Submit the valid T-shirt ID with size=QA-NOT-A-SIZE.
2. Fetch the resulting own order.

Expected result	Actual result
The chosen size belongs to the server-side product variant list.	HTTP 200 success is returned and QA-NOT-A-SIZE is stored and displayed.

Impact The order requests an impossible product variant.

Recommended remedy

Resolve the product from the database and validate the requested variant against its current size list.

Retest acceptance

Unknown sizes fail without persistence; supported sizes succeed.

Evidence

raw/api-validation.json; evidence/orders-validation-tests.json

FM-030 | Order endpoint accepts a non-existent product ID

DEFECT | Data integrity

Module	Order API	Severity / priority	High / P1
Test reference	TC-030	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/order/placeOrder

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Submit item ID 000000000000000000000000 with a copied valid product snapshot.
2. Fetch own orders.

Expected result	Actual result
Every order line references an existing saleable server-side product.	The fabricated ID is accepted and persisted.

Impact An order can reference inventory that does not exist.

Recommended remedy

Look up every product and variant on the server; reject missing or unavailable records before writing the order.

Retest acceptance

Non-existent IDs fail; existing allowed IDs succeed without trusting copied metadata.

Evidence

raw/api-validation.json; raw/products.json

FM-031 | Order endpoint trusts an altered client line-item price

DEFECT | Security

Module	Order API	Severity / priority	High / P1
Test reference	TC-031	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/order/placeOrder

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Copy the valid $200 T-shirt line.
2. Change its client price to 1 and submit amount=11.
3. Fetch own orders.

Expected result	Actual result
The server uses its own catalogue price, ignoring a submitted client price.	The order stores the altered $1 price and $11 amount.

Impact Price manipulation can create underpriced orders. No real payment or fraud was performed.

Recommended remedy

Accept only product ID, variant and quantity; derive immutable line prices and discounts from server rules.

Retest acceptance

Tampered prices are rejected or ignored and the stored price remains $200.

Evidence

raw/api-validation.json

FM-032 | Order endpoint trusts the submitted total amount

DEFECT | Security

Module	Order API	Severity / priority	High / P1
Test reference	TC-032	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/order/placeOrder

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Keep one valid line priced $200 with quantity 1.
2. Submit amount=1 rather than the displayed $210.
3. Fetch own orders.

Expected result	Actual result
The server calculates the payable total from validated lines and shipping.	The amount of 1 is persisted despite the unchanged $200 item.

Impact Totals can be manipulated independently of line-item prices.

Recommended remedy

Recalculate all monetary values on the server with an explicit currency and rounding policy.

Retest acceptance

Altered totals cannot be stored; valid item $200 plus $10 shipping yields $210.

Evidence

raw/api-validation.json

FM-033 | Cart endpoint stores a negative fractional quantity

DEFECT | Data integrity

Module	Cart API	Severity / priority	High / P1
Test reference	TC-033	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/cart/update

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Use only the demo account cart.
2. Update an item quantity to -0.5.
3. Fetch the cart and reload its page.

Expected result	Actual result
Cart quantities remain positive integers; removal uses an explicit operation.	The update succeeds and -0.5 persists. The invalid fixture produced a fractional badge and negative totals.

Impact Cart arithmetic and checkout readiness become invalid.

Recommended remedy

Validate integer bounds server-side and sanitize existing invalid entries before computing totals.

Retest acceptance

Invalid quantities fail and the cart remains unchanged; valid increments and removal still work.

Evidence

raw/cart-validation.json; evidence/session-log.json

Audit note Only the invalid demo entries created for this test were subsequently removed.

FM-034 | Cart endpoint accepts an unsupported size

DEFECT | Data integrity

Module	Cart API	Severity / priority	High / P1
Test reference	TC-034	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/cart/add

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Authenticate as the disposable account.
2. Add the valid product with size=QA-NOT-A-SIZE.
3. Fetch its cart.

Expected result	Actual result
The server rejects a variant not listed for that product.	The unsupported size is stored in cartData.

Impact The cart holds a selection the storefront cannot legitimately fulfil.

Recommended remedy

Resolve the product variant before writing cart state and return a validation error for unknown sizes.

Retest acceptance

Unsupported size is rejected with no cart change; valid size M is accepted.

Evidence

raw/cart-validation.json

Audit note The test entry was removed after evidence capture.

FM-035 | Cart endpoint accepts a non-existent product

DEFECT | Data integrity

Module	Cart API	Severity / priority	High / P1
Test reference	TC-035	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/cart/add

Method Authenticated API test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Add ID 000000000000000000000000, size M to the demo cart.
2. Fetch the cart and reload it.

Expected result	Actual result
The cart accepts only existing saleable products.	The fabricated ID is stored but cannot be resolved into a visible valid product row.

Impact Hidden invalid entries create inconsistent cart state and can break order construction.

Recommended remedy

Validate product existence on write and safely flag or remove stale references on read.

Retest acceptance

Unknown IDs are rejected; deletion or unavailability of a previously valid product has a clear recovery state.

Evidence

raw/cart-validation.json; evidence/session-log.json

Audit note The fabricated test entry was removed after capture.

FM-036 | Guest checkout loses the address when authentication is required

DEFECT | Functionality

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-036	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. As a guest with an item, complete the delivery form.
2. Press PLACE ORDER.
3. Follow the authentication flow.

Expected result	Actual result
Authentication is requested before data entry, or the draft and checkout destination are preserved.	Login is requested only on submission; the entered address is lost and successful signup returns to Home.

Impact A customer must repeat delivery entry and navigate back to checkout.

Recommended remedy

Guard checkout earlier or preserve a draft and returnTo destination across authentication.

Retest acceptance

Guest-to-login and guest-to-signup flows return to checkout with the valid draft and cart intact.

Evidence

evidence/checkout-fields.json; evidence/signup-outcome.json; raw/app.js

FM-037 | Postal field strips letters from an alphanumeric code

DEFECT | Functionality

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-037	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Type SW1A 1AA into Zipcode using normal keystrokes.
2. Read the resulting field value.

Expected result	Actual result
A postal identifier is preserved for a supported destination; unsupported countries are explicitly excluded.	The number input retains only 11 from SW1A 1AA. Country remains unrestricted free text.

Impact A legitimate alphanumeric postal code cannot be entered accurately.

Recommended remedy

Use type=text, preserve spaces and letters, and validate against the supported-country policy.

Retest acceptance

SW1A 1AA is preserved when the UK is supported; otherwise the UI clearly restricts destination selection.

Evidence

evidence/checkout-form-limitations.json; evidence/checkout-form-limitations.jpg

Audit note Supported-country requirements were not supplied; unrestricted country entry makes the current constraint ambiguous.

FM-038 | Phone field rejects a normally formatted telephone string

DEFECT | Functionality

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-038	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Type +1 (415) 555-0321 into Phone number.
2. Inspect the resulting value.

Expected result	Actual result
A telephone string can be entered and normalized without corrupting its meaning.	The type=number control does not retain the entered formatted phone; the recorded value is empty.

Impact Customers cannot paste a commonly formatted contact number.

Recommended remedy

Use type=tel and appropriate inputmode, retaining the original string until validated and normalized.

Retest acceptance

Formatted supported phone numbers work; invalid values receive a field-specific error.

Evidence

evidence/checkout-form-limitations.json; evidence/checkout-form-limitations.jpg

FM-039 | Unknown route has no not-found message or recovery

DEFECT | Functionality

Module	Routing	Severity / priority	Medium / P2
Test reference	TC-039	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/qa-route-does-not-exist

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the stated unknown URL.
2. Allow the application to settle.

Expected result	Actual result
A not-found view explains the invalid route and links to a valid destination.	Only the shared header and footer appear; the route receives the SPA HTML with HTTP 200.

Impact Broken links leave users in an unexplained page and are difficult to diagnose.

Recommended remedy

Add a catch-all route with a clear message and recovery link; use an appropriate status when the hosting architecture permits.

Retest acceptance

Unknown routes show Not found; known deep links still load on refresh.

Evidence

evidence/unknown-route.json; evidence/unknown-route.jpg; raw/route-responses.json

FM-040 | Profile response includes password and refresh-token fields

DEFECT | Security

Module	Account API	Severity / priority	High / P1
Test reference	TC-040	Status	Open
Detected	9 October 2026	Proposed owner	Backend developer

Location https://foreverbackend-k8zm.onrender.com/api/user/getUser

Method Authenticated API response inspection

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Sign in only as the disposable account.
2. Inspect the field names returned by the own-profile endpoint.

Expected result	Actual result
The response contains only client-required profile data and excludes authentication internals.	The field list includes password and refreshToken, as well as internal persistence metadata. Secret values were not copied into this report.

Impact Authentication material is unnecessarily exposed to client code. This is not evidence of access to other users.

Recommended remedy

Return an explicit safe profile DTO and exclude password hashes, refresh tokens and internal metadata from serialization.

Retest acceptance

Own-profile and login responses omit these fields while authorised profile functions still work.

Evidence

raw/api-validation.json

FM-041 | Search opener cannot be activated with the keyboard

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Medium / P2
Test reference	TC-041	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect the search icon.
2. Traverse the header with Tab and attempt keyboard activation.

Expected result	Actual result
The search opener is a named keyboard-operable button.	The clickable image has no button role or tabindex and is absent from the normal interactive sequence.

Impact Keyboard-only shoppers cannot open search.

Recommended remedy

Use a button with an accessible name such as Open search, preserving the icon as decorative.

Retest acceptance

Tab reaches the opener and Enter or Space opens the search form.

Evidence

evidence/home-mobile.json; raw/app.js

FM-042 | Search close icon lacks a keyboard action and name

DEFECT | Accessibility

Module	Collection	Severity / priority	Medium / P2
Test reference	TC-042	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the search bar.
2. Inspect its close icon and try keyboard traversal.

Expected result	Actual result
Close search is a named button reachable with Tab.	The close affordance is a clickable image with empty alternative text and no button semantics.

Impact The visible closing action is unavailable to keyboard and screen-reader users.

Recommended remedy

Wrap the icon in a button labelled Close search and restore focus to the opener when it closes.

Retest acceptance

The action is announced by name and closes the bar with Enter or Space.

Evidence

evidence/search-no-results.json; raw/app.js

FM-043 | Authenticated account menu is not keyboard operable

DEFECT | Accessibility

Module	Account	Severity / priority	Medium / P2
Test reference	TC-043	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Sign in and inspect the profile opener.
2. Inspect Orders and Logout in its menu.

Expected result	Actual result
The opener and menu actions have suitable keyboard behaviour and semantic roles.	The opener is a clickable div and image; menu actions are clickable paragraphs without normal keyboard semantics.

Impact Keyboard users cannot reliably open or use the account menu, including logout.

Recommended remedy

Use a named disclosure button, a link for Orders and a button for Logout with proper focus handling.

Retest acceptance

The entire account menu can be opened, traversed, activated and dismissed using the keyboard.

Evidence

evidence/profile-no-action.json; raw/app.js

FM-044 | Login and Create Account mode switch is not a keyboard control

DEFECT | Accessibility

Module	Account	Severity / priority	Medium / P2
Test reference	TC-044	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/login

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the account form.
2. Inspect and attempt keyboard activation of Login or Create Account.

Expected result	Actual result
The mode switch is an operable button or link.	The switch is a paragraph with an onClick handler and no keyboard control semantics.

Impact Keyboard users cannot select the alternative authentication form.

Recommended remedy

Use a type=button mode switch or links to explicit account routes.

Retest acceptance

Enter and Space switch modes without submitting the form accidentally.

Evidence

evidence/login-form.json; raw/app.js

FM-045 | Product gallery thumbnails are pointer-only controls

DEFECT | Accessibility

Module	Product	Severity / priority	Medium / P2
Test reference	TC-045	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the product with multiple gallery images.
2. Inspect the thumbnail controls.

Expected result	Actual result
Each thumbnail is keyboard operable and identifies the image it selects.	Thumbnails are clickable images without button roles or tabindex.

Impact Keyboard shoppers cannot inspect alternative product views.

Recommended remedy

Use named buttons around thumbnails, exposing selected state and a visible focus indicator.

Retest acceptance

All images can be selected using Tab and Enter; selected image state is announced.

Evidence

evidence/product-details.json; raw/app.js

FM-046 | Remove cart item is an unnamed pointer-only image

DEFECT | Accessibility

Module	Cart	Severity / priority	Medium / P2
Test reference	TC-046	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/cart

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Add one valid item.
2. Inspect the bin icon used to remove the row.

Expected result	Actual result
Removal is a named keyboard-operable button identifying its item.	The bin is an image click target without suitable alternative text or button semantics.

Impact Keyboard and assistive-technology users cannot reliably remove a line item.

Recommended remedy

Use a button labelled Remove [product name] with a clear focus style.

Retest acceptance

The item is removable with the keyboard and focus moves to a sensible remaining control.

Evidence

evidence/cart-one-item.json; raw/app.js

FM-047 | Cart product navigation is available only through a clickable image

DEFECT | Accessibility

Module	Cart	Severity / priority	Medium / P2
Test reference	TC-047	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/cart

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Add one item.
2. Inspect its image-based route to the product.

Expected result	Actual result
An ordinary product link is available and identifies the destination.	The image uses a click handler rather than an anchor; the visible product name is also blank.

Impact Keyboard users cannot use the cart shortcut back to product details.

Recommended remedy

Use an anchor containing an informative product name or image alternative.

Retest acceptance

Tab reaches a named product link and Enter opens the correct product.

Evidence

evidence/cart-one-item.json; raw/app.js

FM-048 | Mobile menu opener has no button semantics

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Medium / P2
Test reference	TC-048	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Responsive DOM review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Use a 390 px viewport.
2. Inspect the menu icon.

Expected result	Actual result
Open menu is a named button with its expanded state exposed.	The visible opener is a clickable image without button semantics or aria-expanded.

Impact Keyboard and assistive-technology users cannot reliably open mobile navigation.

Recommended remedy

Use a button with aria-expanded and aria-controls tied to the menu panel.

Retest acceptance

Keyboard activation opens the menu and the announced state changes.

Evidence

evidence/home-mobile.json; evidence/mobile-menu-open.json; raw/app.js

FM-049 | Mobile menu Back affordance is not keyboard operable

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Medium / P2
Test reference	TC-049	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Responsive DOM review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open mobile navigation.
2. Inspect its Back image and text.

Expected result	Actual result
The visible closing control is a keyboard-operable, named button.	Back is implemented with a clickable div, image and paragraph rather than a button.

Impact A keyboard user lacks a reliable explicit close control.

Recommended remedy

Use a Close menu button and return focus to the original opener.

Retest acceptance

Tab reaches Close menu; Enter or Space closes the panel and restores focus.

Evidence

evidence/mobile-menu-open.json; raw/app.js

FM-050 | Closed mobile menu leaves offscreen links in the accessibility tree

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Medium / P2
Test reference	TC-050	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Responsive DOM and code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. At mobile width, close the menu.
2. Inspect the menu links and their hiding mechanism.

Expected result	Actual result
Collapsed navigation content is removed from interactive and accessibility navigation.	The panel is moved with translate-x-full; its links remain in the snapshot and are not inert or aria-hidden.

Impact Assistive navigation can expose offscreen destinations unrelated to the visible interface.

Recommended remedy

Conditionally mount the closed panel or use an appropriate inert/hidden state, restoring it when opened.

Retest acceptance

Closed links are absent from the accessibility and tab sequence; opening makes them available.

Evidence

evidence/home-mobile.json; raw/app.js

Audit note The observed accessibility-tree exposure is confirmed; individual assistive-technology behaviour was not separately tested.

FM-051 | Mobile menu does not manage focus or dismiss on Escape

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Medium / P2
Test reference	TC-051	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the mobile menu.
2. Press Escape while a navigation item is focused.
3. Inspect focus handling.

Expected result	Actual result
An overlay menu provides predictable focus and dismissal behaviour.	Escape leaves the menu open; no opening focus transfer or closing focus restoration is implemented.

Impact Keyboard users can lose their position in navigation.

Recommended remedy

Choose a disclosure or dialog interaction model and implement its focus and Escape behaviour consistently.

Retest acceptance

Open, traverse, Escape and reopen the menu; focus remains predictable throughout.

Evidence

evidence/mobile-menu-escape.json; evidence/mobile-menu-open.json; raw/app.js

FM-052 | Mobile filter disclosure is a paragraph rather than a control

DEFECT | Accessibility

Module	Collection	Severity / priority	Medium / P2
Test reference	TC-052	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method Responsive DOM review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Use mobile Collection.
2. Inspect the FILTERS toggle.

Expected result	Actual result
A button opens filters and exposes expanded or collapsed state.	The toggle is a clickable paragraph without button semantics or expanded-state metadata.

Impact Keyboard and assistive users cannot reliably discover and operate filtering.

Recommended remedy

Use a button with aria-expanded and aria-controls for the filter group.

Retest acceptance

Keyboard opening works and the state is announced correctly.

Evidence

evidence/mobile-filters-open.json; raw/app.js

FM-053 | Payment methods are not keyboard selectable radio controls

DEFECT | Accessibility

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-053	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method DOM and public code review

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Open checkout.
2. Traverse to the Stripe and Cash on Delivery choices.

Expected result	Actual result
Payment methods form a labelled radio group with keyboard selection and selected-state exposure.	The choices are clickable divs and decorative circles, with no radio or button semantics.

Impact Keyboard customers cannot select a payment method reliably.

Recommended remedy

Use labelled radio inputs in a fieldset and expose the selected payment option.

Retest acceptance

Tab and arrow keys select each method; the selected method is announced.

Evidence

evidence/checkout-form-limitations.json; raw/app.js

FM-054 | Stripe option has no accessible payment-provider name

DEFECT | Accessibility

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-054	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method DOM review

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Inspect the Stripe choice in the checkout accessibility snapshot.

Expected result	Actual result
The provider name Stripe is available as text or an accessible label.	Only a logo with alt="" appears; the option has no text label and is absent as a named choice.

Impact A screen-reader user cannot identify this visible payment option.

Recommended remedy

Add the text label Stripe to its radio option; retain the logo as decorative.

Retest acceptance

The provider is announced as Stripe together with its selected state.

Evidence

evidence/checkout-form-limitations.json; evidence/checkout-form-limitations.jpg

Audit note This checks the accessible name, separately from the keyboard-selection defect.

FM-055 | Selected product size is communicated only by colour

DEFECT | Accessibility

Module	Product	Severity / priority	Medium / P2
Test reference	TC-055	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Select size M.
2. Inspect its button attributes before and after selection.

Expected result	Actual result
The selected variant has a programmatically exposed state.	Only the border/background classes change; no aria-pressed or radio checked state indicates the selected size.

Impact Assistive users cannot determine which size will be added.

Recommended remedy

Use a labelled size radio group, or expose aria-pressed consistently on selection buttons.

Retest acceptance

Selecting M announces its state and clears the previous selection.

Evidence

evidence/related-size-carryover.json; raw/app.js

FM-056 | Collection sort selector has no associated label

DEFECT | Accessibility

Module	Collection	Severity / priority	Medium / P2
Test reference	TC-056	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method DOM and code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect the sort select element and its labels or aria-label.

Expected result	Actual result
The control has a programmatic label explaining its purpose.	The select has no label, id association or aria-label; only option text contains Sort by.

Impact The control purpose depends on selected option text rather than a stable accessible name.

Recommended remedy

Associate a Sort products label with the select.

Retest acceptance

The control is announced by its purpose with each available option.

Evidence

evidence/collection-loaded.jpg; raw/app.js

FM-057 | Delivery fields rely on disappearing placeholders

DEFECT | Accessibility

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-057	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Inspect delivery inputs before entry.
2. Type values into first name, address and contact fields.

Expected result	Actual result
Persistent visible labels identify the fields after entry and provide programmatic associations.	There are no associated labels; placeholders disappear as values are entered. Numeric postal and phone controls appear unnamed in the snapshot.

Impact Customers lose field context and assistive users encounter unnamed controls.

Recommended remedy

Add label elements linked to stable input IDs for every delivery field; keep placeholders as optional examples.

Retest acceptance

All nine fields retain visible and accessible names after entry.

Evidence

evidence/checkout-fields.json; evidence/checkout-form-limitations.json

FM-058 | Authentication fields have no persistent associated labels

DEFECT | Accessibility

Module	Account	Severity / priority	Medium / P2
Test reference	TC-058	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/login

Method DOM and public code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open Signup or Login.
2. Fill username, email and password and inspect labels.

Expected result	Actual result
Each credential field retains a visible label and programmatic association.	Only placeholders identify the fields; no label associations are present.

Impact Entered forms lose clear visual context and depend on placeholder interpretation.

Recommended remedy

Add associated Username, Email and Password labels to the account form.

Retest acceptance

Both modes keep field names visible and expose them correctly after entry.

Evidence

evidence/login-form.json; raw/app.js

FM-059 | Personal-data inputs do not expose their autocomplete purposes

DEFECT | Accessibility

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-059	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method DOM review

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Inspect autocomplete attributes on delivery name, address, email and phone inputs.

Expected result	Actual result
Known personal-data purposes are exposed using appropriate autocomplete tokens.	The fields do not specify purpose tokens such as given-name, street-address, postal-code or tel.

Impact Assistive personalization and reliable browser autofill lack explicit field-purpose metadata.

Recommended remedy

Add correct autocomplete tokens for each supported personal-data field and authentication mode.

Retest acceptance

Inspect the corrected DOM and verify a test autofill preserves each field purpose.

Evidence

evidence/checkout-form-limitations.json; evidence/login-form.json

FM-060 | Muted home-page text has insufficient contrast

DEFECT | Accessibility

Module	Home	Severity / priority	Medium / P2
Test reference	TC-060	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Computed-style measurement

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect policy and newsletter text rendered in gray-400 on white.
2. Calculate contrast from the observed computed colour.

Expected result	Actual result
Normal-size text meets a minimum 4.5:1 contrast ratio.	The computed colour is oklch(0.707 0.022 261.325) at 16 px on white, approximately RGB [153, 161, 175]. The calculated contrast ratio is 2.60:1, below 4.5:1.

Impact Low-vision users may struggle to read service-policy and newsletter text.

Recommended remedy

Use a darker approved text colour and check all states against the real background.

Retest acceptance

The same text at 16 px measures at least 4.5:1 without relying on rounding.

Evidence

evidence/typography-semantics.json; evidence/contrast-measurement.json

Audit note The exact converted sRGB colour and ratio are included in the evidence summary.

FM-061 | Newsletter input suppresses its visible keyboard focus

DEFECT | Accessibility

Module	Home	Severity / priority	Medium / P2
Test reference	TC-061	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Keyboard test and style review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Focus SUBSCRIBE, then Shift+Tab into the email input.
2. Inspect the focus outline and screenshot.

Expected result	Actual result
Keyboard focus is visibly indicated on the input.	The input uses outline-none and no focus shadow; the observed focus has no visible outline or replacement.

Impact Keyboard users cannot easily locate the active subscription field.

Recommended remedy

Provide a distinct focus-visible border or outline with sufficient contrast.

Retest acceptance

Tab and Shift+Tab clearly indicate focus without requiring pointer interaction.

Evidence

evidence/newsletter-focus.jpg; raw/app.js

FM-062 | Meaningful product photographs have empty alternative text

DEFECT | Accessibility

Module	Product	Severity / priority	Medium / P2
Test reference	TC-062	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/product/68ee7c73675d613eba9e96c6

Method DOM review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect the main product image and gallery alternatives.

Expected result	Actual result
Meaningful product views have concise text alternatives describing the item or view.	The product images use alt="" despite conveying garment identity and appearance.

Impact Screen-reader users cannot obtain information conveyed by the product photography.

Recommended remedy

Provide accurate item and view alternatives; use empty alt only for genuinely decorative or fully redundant images.

Retest acceptance

The main image and each informative gallery view expose useful, non-duplicative alternatives.

Evidence

evidence/product-details.json; raw/app.js

FM-063 | Visible page and section titles are not semantic headings

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Medium / P2
Test reference	TC-063	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method DOM review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect Collection, Cart, Checkout, About and Contact heading outlines.

Expected result	Actual result
Page titles and major sections use a logical h1-h2 hierarchy.	The inspected pages have paragraph-based titles and no h1-h3 outline; the common title component renders a p.

Impact Assistive heading navigation cannot follow the visible document structure.

Recommended remedy

Use semantic headings in the shared title component with context-appropriate levels.

Retest acceptance

Every page has a meaningful h1 and logical section headings without skipped structural levels.

Evidence

evidence/collection-reload.json; evidence/about-page.json; evidence/contact-page.json; raw/app.js

FM-064 | Page content lacks main and footer landmarks

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Low / P3
Test reference	TC-064	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method DOM review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect document landmarks on Home and Collection.

Expected result	Actual result
Main content and the site footer have suitable landmark semantics.	No main or contentinfo/footer landmarks are present; major regions are generic containers.

Impact Assistive users have fewer efficient ways to navigate page regions.

Recommended remedy

Use one main element per page and a footer for shared closing content.

Retest acceptance

Landmark navigation identifies Navigation, Main and Content information in a sensible order.

Evidence

evidence/typography-semantics.json

FM-065 | Browser title does not identify the current page

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Low / P3
Test reference	TC-065	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Navigate between Home, Collection, Product, About and Checkout.
2. Compare document.title.

Expected result	Actual result
Each title describes its page purpose and the store.	All inspected routes retain Forever Mart.

Impact Tabs and assistive page announcements do not distinguish destinations.

Recommended remedy

Set route-specific titles such as Collection | ForeverMart and include product names on detail pages.

Retest acceptance

Each route and direct deep link exposes an accurate title.

Evidence

evidence/collection-reload.json; evidence/product-details.json; evidence/about-page.json; evidence/checkout-fields.json

FM-066 | Layout overflows horizontally at 320 CSS pixels

DEFECT | Accessibility

Module	Shared layout	Severity / priority	Medium / P2
Test reference	TC-066	Status	Open
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Responsive measurement

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Set the viewport to 320 CSS px.
2. Inspect the header and checkout document width.

Expected result	Actual result
The page reflows without horizontal scrolling for ordinary content.	At 320 CSS px, the settled checkout document and body are both 352 px wide. The totals panel uses min-w-80 within the page margins.

Impact Small-screen and zoomed users must pan horizontally to reach content.

Recommended remedy

Remove fixed minimum widths that exceed the available content area; wrap header controls and stacked form panels.

Retest acceptance

At 320 CSS px, document width is no greater than the viewport and all controls remain usable.

Evidence

evidence/checkout-320-verified.json; evidence/checkout-320-verified.jpg; raw/app.js

Audit note Reproduced again at the stated viewport; a screenshot and the full DOM snapshot are retained.

FM-067 | Home-page hero misspells Arrivals

DEFECT | Content

Module	Home	Severity / priority	Low / P3
Test reference	TC-067	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open Home and read the hero heading.

Expected result	Actual result
The heading reads Latest Arrivals.	The visible heading reads Latest Arivals.

Impact The first impression contains a spelling mistake.

Recommended remedy

Correct the copy and add a content proofreading checklist.

Retest acceptance

The rendered heading and accessible heading text both read Latest Arrivals.

Evidence

evidence/home-desktop.jpg; raw/app.js

FM-068 | Template filler text remains in customer-facing sections

DEFECT | Content

Module	Shared layout	Severity / priority	Low / P3
Test reference	TC-068	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Read the home collection descriptions, newsletter copy and footer.

Expected result	Actual result
Public copy explains the store and each section.	Several visible paragraphs contain Lorem ipsum template text.

Impact Customers see unfinished content instead of useful information.

Recommended remedy

Replace all placeholder paragraphs with approved store copy and search the production content for template text.

Retest acceptance

No Lorem ipsum remains on public routes; reviewed copy is meaningful.

Evidence

evidence/home-desktop.jpg; evidence/about-page.json; raw/app.js

FM-069 | Contact details conflict between the footer and Contact page

DEFECT | Content

Module	Contact	Severity / priority	Medium / P2
Test reference	TC-069	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/contact

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Compare Contact page details with its footer.

Expected result	Actual result
The store publishes one verified contact set, or clearly labels different departments.	The footer shows +1-212-444-780 and contact@foreveryou.com; Contact shows (415) 555-0321 and admin@forever.com without an explanation.

Impact Customers do not know which contact details belong to the store.

Recommended remedy

Centralize approved contact data and label any distinct support or administrative contacts.

Retest acceptance

Each route shows the same approved support contact or explains the difference.

Evidence

evidence/contact-page.json; evidence/contact-page.jpg

FM-070 | About page contains an incomplete support sentence

DEFECT | Content

Module	About	Severity / priority	Low / P3
Test reference	TC-070	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/about

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Read the support paragraph on About.

Expected result	Actual result
The sentence is complete and understandable.	The copy ends with assist you the way, leaving the intended meaning incomplete.

Impact The About page communicates unfinished copy.

Recommended remedy

Rewrite the sentence and proofread the complete About page.

Retest acceptance

The paragraph is grammatically complete and approved by the content owner.

Evidence

evidence/about-page.json

FM-071 | Winterwear filter has no matching catalogue classification

OBSERVATION | Usability

Module	Collection	Severity / priority	Low / P3
Test reference	TC-071	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Select winterwear and compare the public subCategory values.

Expected result	Actual result
Filter options reflect available classifications or explain an empty category.	winterwear returns an empty grid; the catalogue uses only Topwear and Bottomwear, including for jackets.

Impact An apparently useful filter has no compatible records.

Recommended remedy

Align the taxonomy and product classification, or omit unavailable options until they contain products.

Retest acceptance

Every advertised filter is backed by the intended classification or has an explicit empty state.

Evidence

evidence/winterwear-filter.json; raw/products.json

Audit note The intended garment taxonomy requires owner confirmation.

FM-072 | Empty cart has no explanatory state or shopping recovery

OBSERVATION | Usability

Module	Cart	Severity / priority	Low / P3
Test reference	TC-072	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/cart

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Open the cart with zero items.

Expected result	Actual result
The page states that the cart is empty and offers a clear shopping action.	There are no rows or empty-cart explanation, but checkout remains the main action.

Impact Customers receive little guidance when nothing is selected.

Recommended remedy

Add an empty-cart message and a Continue shopping link; suppress checkout in that state.

Retest acceptance

A new or cleared cart has a clear recovery action.

Evidence

evidence/cart-empty.json; evidence/cart-empty.jpg

FM-073 | Checkout does not show the products being purchased

OBSERVATION | Usability

Module	Checkout	Severity / priority	Medium / P2
Test reference	TC-073	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Open checkout after adding a valid item.
2. Inspect the review information.

Expected result	Actual result
A shopper can verify product, variant and quantity before submission.	Only monetary totals and address fields are shown; no item summary is presented.

Impact The final screen does not help customers catch selection mistakes.

Recommended remedy

Include a concise order summary with item names, quantities, sizes and prices, with an edit-cart link.

Retest acceptance

The summary agrees with the cart and server-calculated final order.

Evidence

evidence/checkout-fields.json; evidence/checkout-fields.jpg

FM-074 | Order history omits order identifiers

OBSERVATION | Usability

Module	Orders	Severity / priority	Medium / P2
Test reference	TC-074	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/order

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Open own demo order history and compare it with the own-orders API.

Expected result	Actual result
Customers have a visible order reference for support.	The API has order IDs, but the interface does not display them.

Impact A customer cannot cite an unambiguous order reference from the interface.

Recommended remedy

Show a safe order reference on each order group and use it in support or detail links.

Retest acceptance

Each displayed order has a unique visible reference matching its record.

Evidence

evidence/orders-validation-tests.json; raw/api-validation.json

FM-075 | Order history loses order-level grouping

OBSERVATION | Usability

Module	Orders	Severity / priority	Medium / P2
Test reference	TC-075	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/order

Method Browser and public code review

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Inspect the own order history and the public order-view component.

Expected result	Actual result
Line items are grouped under the order that owns them.	The component pushes all order items into one list and renders item rows without order containers.

Impact Customers cannot clearly distinguish separate purchases with similar items.

Recommended remedy

Render order-level sections containing their line items, reference, date and total.

Retest acceptance

Two orders and a multi-item order are visually distinguishable; empty invalid records are not silently hidden.

Evidence

evidence/orders-validation-tests.json; raw/app.js

Audit note The flattening implementation was inspected; a multi-item-order scenario was not executed.

FM-076 | Order history omits the final order total and shipping breakdown

OBSERVATION | Usability

Module	Orders	Severity / priority	Medium / P2
Test reference	TC-076	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/order

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Inspect the order priced $210 in the own-orders data and its visible row.

Expected result	Actual result
The customer can verify the stored final payable amount.	Only item price and quantity are shown; the stored order total and shipping breakdown are not displayed.

Impact The view cannot serve as a complete order cost confirmation.

Recommended remedy

Show order subtotal, shipping, discounts where applicable, currency and final total from the validated order record.

Retest acceptance

The $200 test item plus $10 shipping displays a $210 final total.

Evidence

evidence/orders-validation-tests.json; raw/api-validation.json

FM-077 | Order history has no delivery-address review

OBSERVATION | Usability

Module	Orders	Severity / priority	Low / P3
Test reference	TC-077	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/order

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Open own order history after the test order.
2. Inspect the delivery information available to the customer.

Expected result	Actual result
An order detail view allows the customer to review the submitted delivery address safely.	The visible history provides no address or detail link for reviewing it.

Impact A delivery-entry mistake cannot be checked from the current customer interface.

Recommended remedy

Provide an authenticated order-detail view with the delivery address and contact information.

Retest acceptance

Own order details show the stored address; another account cannot retrieve it.

Evidence

evidence/orders-validation-tests.json; raw/demo-orders.json

FM-078 | About page advertises product ranges absent from the inspected catalogue

OBSERVATION | Content

Module	About	Severity / priority	Low / P3
Test reference	TC-078	Status	Review required
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/about

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Read the advertised product ranges.
2. Compare with the 40 returned product records.

Expected result	Actual result
Marketing copy reflects the current offer or explicitly describes future ranges.	About mentions electronics, beauty and home essentials while the inspected catalogue contains apparel only.

Impact The current copy can create expectations the store does not presently meet.

Recommended remedy

Align copy with the current catalogue or label future expansion clearly.

Retest acceptance

Each current-range claim is supported by published products.

Evidence

evidence/about-page.json; raw/products.json

FM-079 | Newsletter placeholder contains incorrect wording

DEFECT | Content

Module	Home	Severity / priority	Low / P3
Test reference	TC-079	Status	Open
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect the newsletter email placeholder.

Expected result	Actual result
The field instruction is clear and correctly written.	The placeholder reads enter you email.

Impact A visible entry instruction contains a grammar error.

Recommended remedy

Use a persistent Email address label and an optional example placeholder.

Retest acceptance

The form contains clear approved wording before and after entry.

Evidence

evidence/home-mobile.json

FM-080 | Copyright footer contains incomplete grammar and inconsistent branding

OBSERVATION | Content

Module	Shared layout	Severity / priority	Low / P3
Test reference	TC-080	Status	Review required
Detected	9 October 2026	Proposed owner	Content owner

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Read the copyright footer and compare its named domain with the store identity.

Expected result	Actual result
The notice is correctly written and uses the approved store or legal-owner identity.	The line reads Copyright 2025@ forever.com -All Right Reserved.; the store is presented as Forever Mart.

Impact The footer appears unfinished and its ownership identity needs clarification.

Recommended remedy

Use the approved legal identity and All rights reserved wording. Verify the year policy rather than automatically replacing it.

Retest acceptance

The notice is grammatically correct and its owner is approved.

Evidence

evidence/home-mobile.json

Audit note A 2025 year is not itself classified as a defect; it may identify the original publication year.

FM-081 | Public document has no Content Security Policy

OBSERVATION | Security

Module	Platform	Severity / priority	Medium / P2
Test reference	TC-081	Status	Review required
Detected	9 October 2026	Proposed owner	Platform developer

Location https://forevermart.vercel.app/

Method HTTP configuration review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect the saved response headers and public HTML head.

Expected result	Actual result
A production hardening policy limits script and other resource sources as appropriate to this app.	Neither a Content-Security-Policy response header nor a CSP meta policy is present in the inspected document.

Impact The page lacks this defence-in-depth control; no script-injection exploit was demonstrated.

Recommended remedy

Develop a least-privilege CSP using report-only validation before enforcing it, including the actual API, image and payment dependencies.

Retest acceptance

The reviewed policy is present and all authorised app flows work without unexpected violations.

Evidence

raw/headers.json; raw/home.html

FM-082 | Public document lacks an explicit framing restriction

OBSERVATION | Security

Module	Platform	Severity / priority	Medium / P2
Test reference	TC-082	Status	Review required
Detected	9 October 2026	Proposed owner	Platform developer

Location https://forevermart.vercel.app/

Method HTTP configuration review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect frame-related response headers and policy directives.

Expected result	Actual result
The deployment restricts embedding to explicitly approved origins.	No X-Frame-Options or CSP frame-ancestors restriction is present in the inspected response.

Impact Clickjacking defence is not configured in this response; an exploit was not attempted.

Recommended remedy

Set a suitable CSP frame-ancestors policy and, where appropriate, X-Frame-Options for older clients.

Retest acceptance

An unauthorised framing origin is blocked and approved integrations continue to work.

Evidence

raw/headers.json

FM-083 | Public responses omit the nosniff hardening header

OBSERVATION | Security

Module	Platform	Severity / priority	Low / P3
Test reference	TC-083	Status	Review required
Detected	9 October 2026	Proposed owner	Platform developer

Location https://forevermart.vercel.app/

Method HTTP configuration review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect X-Content-Type-Options in the saved public response.

Expected result	Actual result
Responses use correct content types and X-Content-Type-Options: nosniff.	The inspected response contains no X-Content-Type-Options header.

Impact An additional browser MIME-sniffing defence is absent; no MIME-confusion exploit was demonstrated.

Recommended remedy

Configure nosniff consistently and verify asset MIME types.

Retest acceptance

HTML, script, stylesheet and image responses have correct types and the intended header.

Evidence

raw/headers.json

FM-084 | Public document has no explicit Referrer Policy

OBSERVATION | Security

Module	Platform	Severity / priority	Low / P3
Test reference	TC-084	Status	Review required
Detected	9 October 2026	Proposed owner	Platform developer

Location https://forevermart.vercel.app/

Method HTTP configuration review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect Referrer-Policy headers and meta tags in the document.

Expected result	Actual result
The application explicitly defines the referrer information it permits external destinations to receive.	No explicit policy is present; browser defaults determine behaviour.

Impact The deployment leaves this privacy setting implicit; this finding does not claim a proven data leak.

Recommended remedy

Choose and configure an appropriate policy such as strict-origin-when-cross-origin after reviewing navigation needs.

Retest acceptance

Inspect same-origin and cross-origin navigation to confirm the intended information is sent.

Evidence

raw/headers.json; raw/home.html

FM-085 | Offscreen catalogue images are not marked for lazy loading

OBSERVATION | Performance

Module	Collection	Severity / priority	Low / P3
Test reference	TC-085	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method DOM and code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Load the 40-product collection.
2. Inspect loading attributes on product images below the fold.

Expected result	Actual result
Non-critical offscreen images can defer loading while initial visible content remains prioritised.	The inspected image elements have no loading=lazy attribute.

Impact The implementation misses a possible transfer-saving optimisation; no Core Web Vitals failure is claimed.

Recommended remedy

Lazy-load offscreen cards, keep initial visible images eager, and measure the impact on a representative device.

Retest acceptance

A network trace verifies deferred offscreen requests without delaying visible products.

Evidence

evidence/collection-reload.json; evidence/typography-semantics.json; raw/app.js

FM-086 | Product images do not advertise responsive source variants

OBSERVATION | Performance

Module	Collection	Severity / priority	Low / P3
Test reference	TC-086	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method DOM review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect catalogue image srcset and sizes attributes at desktop and mobile widths.

Expected result	Actual result
Image delivery can select a suitably sized source for the rendered card.	The inspected images use a single source with no srcset or sizes attributes.

Impact Smaller screens cannot request a deliberately smaller variant through these attributes; actual speed impact was not measured.

Recommended remedy

Generate appropriate Cloudinary or equivalent variants and declare sizes and srcset.

Retest acceptance

Mobile and desktop request suitable image dimensions while retaining image quality.

Evidence

evidence/typography-semantics.json; evidence/collection-mobile.json

FM-087 | Public pages lack a page-specific meta description

OBSERVATION | Discoverability

Module	Routing	Severity / priority	Low / P3
Test reference	TC-087	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/collection

Method HTML and code review

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect the document head and route title updates.

Expected result	Actual result
Indexable store pages provide concise, page-specific descriptive metadata.	The inspected HTML has no meta description and the client does not populate a route-specific description.

Impact Search and sharing systems have little authored summary information. This is an optimisation observation.

Recommended remedy

Add approved descriptions for indexable routes and product pages using a route-aware metadata strategy.

Retest acceptance

Rendered page heads contain distinct relevant descriptions without indexing private account routes.

Evidence

raw/home.html; evidence/typography-semantics.json; raw/app.js

FM-088 | Footer email and telephone cannot be activated directly

OBSERVATION | Usability

Module	Contact	Severity / priority	Low / P3
Test reference	TC-088	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/contact

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Inspect and click the email and telephone strings in the footer and contact details.

Expected result	Actual result
Customers can activate an approved support email or phone link on compatible devices.	The details are static text without mailto or tel links.

Impact Customers must manually copy contact information, especially awkward on phones.

Recommended remedy

After approving the contact values, provide mailto and tel links with meaningful accessible names.

Retest acceptance

The links invoke the corresponding device action without sending a message automatically.

Evidence

evidence/contact-page.json; raw/app.js

FM-089 | Checkout does not state which dollar currency is used

OBSERVATION | Usability

Module	Checkout	Severity / priority	Low / P3
Test reference	TC-089	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/PlaceOrder

Method Browser test

Preconditions Authenticated disposable demo account; cash on delivery; no real payment.

Reproduction procedure

1. Inspect the displayed prices and final checkout total.

Expected result	Actual result
The payable amount identifies its currency unambiguously.	Only $ is shown; no ISO currency label is displayed while the country field is unrestricted.

Impact An international customer cannot distinguish USD from other dollar currencies from the checkout display.

Recommended remedy

Display the actual supported ISO currency next to the final total and align the payment configuration.

Retest acceptance

Cart, checkout and order confirmation use one consistent, explicit currency.

Evidence

evidence/checkout-fields.json; raw/app.js

FM-090 | Displayed return and exchange promises lack instructions

OBSERVATION | Usability

Module	Home	Severity / priority	Low / P3
Test reference	TC-090	Status	Review required
Detected	9 October 2026	Proposed owner	Frontend developer

Location https://forevermart.vercel.app/

Method Browser test

Preconditions Public deployment available; use the stated page and test data.

Reproduction procedure

1. Read Easy Exchange Policy and 7 Days Return Policy.
2. Look for a working route to the policy details.

Expected result	Actual result
Customers can read the scope, conditions and procedure behind these service promises.	The promises are brief text only; the inspected interface provides no usable policy-detail route or procedure.

Impact Customers cannot determine how to request the promised service.

Recommended remedy

Publish owner-approved policy details and link directly from the service summary.

Retest acceptance

A customer can find eligibility, timeframe, procedure and support contact from each summary.

Evidence

evidence/home-mobile.json; raw/app.js

Requirement and defect traceability

These are audit criteria, not a claim that a formal requirements specification was supplied. Every detailed record maps to a test with the same numeric suffix; TC-031, for example, reproduces FM-031.

Criterion	Test and finding references	Expected control
AC-01 Store actions	FM-001, 004, 015-019	Advertised actions open or perform their declared function
AC-02 Product truth	FM-002, 005-011, 020	Product identity, ratings, variants and names are accurate
AC-03 Find and retain choices	FM-012-014, 021, 036	Search normalization and state transitions preserve intent
AC-04 Deliverable order	FM-022, 024-030, 037-038	Valid non-empty items and usable delivery data
AC-05 Price integrity	FM-023, 031-032	Server-derived prices, shipping and totals reconcile
AC-06 Valid cart	FM-033-035	Existing product, valid variant, positive integer quantity
AC-07 Safe profile data	FM-040	Exclude authentication internals from client responses
AC-08 Accessible operation	FM-041-066	Keyboard semantics, labels, state, contrast and reflow
AC-09 Accurate public copy	FM-067-070, 079	Reviewed text and consistent contacts
AC-10 Product-owner review	FM-003, 071-078, 080-090	Decide and approve observations before treating them as defects

Detailed findings contain the exact expected-result assertion and proposed retest. Adapted from Lewis E6 test-case fields, p. 358 (PDF p. 385), and E29 traceability reporting, p. 384 (PDF p. 411).

Checkout decision table

The table states the proposed business rules and the actual checked outcomes. A dash means the condition is irrelevant to that row; the table does not imply that every combination was executed.

Case	Signed in	Valid items	Valid address	Trusted total	Expected action	Observed
DT-01	Yes	No, empty	Yes	-	Reject	Accepted: FM-022
DT-02	Yes	No, qty -0.5	Yes	Yes	Reject	Accepted: FM-028
DT-03	Yes	No, size unknown	Yes	Yes	Reject	Accepted: FM-029
DT-04	Yes	No, ID unknown	Yes	Yes	Reject	Accepted: FM-030
DT-05	Yes	Yes	No, spaces	-	Reject	Accepted: FM-024
DT-06	Yes	Yes	No, email	Yes	Reject	Accepted: FM-027
DT-07	Yes	Yes	Yes	No, altered	Recalculate or reject	Accepted: FM-031/032
DT-08	Yes	Yes	Yes	Yes	Allow COD order	Baseline demo order works

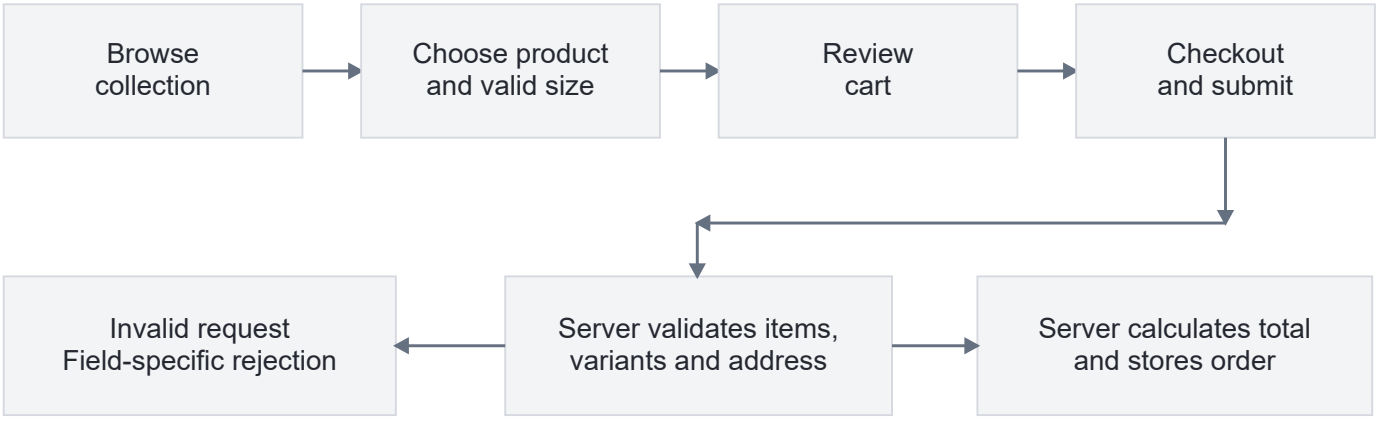
Boundary partitions for the next iteration

Field	Valid partition	Invalid partitions to retest
Quantity	Positive integer within maximum	Negative, zero, fraction, non-number, over maximum
Product and size	Existing saleable product and supported size	Unknown ID, unavailable product, unsupported size
Address	Trimmed required strings and country-appropriate contact identifiers	Whitespace-only, malformed email/phone/postal code
Money	Server catalogue and authorised adjustments	Client price, total, shipping or currency changes

Project-specific adaptation of Lewis G11 Decision Tables and Exhibit G.35, p. 492 (PDF p. 519).

Transaction workflow and control boundary

The inspected application loads public product data, maintains cart state and submits a delivery address plus copied item data to its API. The validation and pricing results show why the trust boundary must be enforced on the server.



Proposed correction: perform validation and pricing at the server boundary.

Figure 1 Proposed purchase validation workflow. This is a corrective design, not a claim that these controls already exist.

Boundary	Observed issue	Required remedy
Client to order API	Altered line price and submitted total persist	Resolve server product records and calculate final money
Client to cart API	Unknown IDs/sizes and quantity -0.5 persist	Validate identity, variant and quantity before writing
Client form to order API	Spaces, -1 contact data and malformed email persist	Use one server-side address schema
API to browser profile	password and refreshToken fields returned	Serialize an allowlisted client profile object

Flowchart method adapted from Lewis Chapter 3, p. 42 (PDF p. 69). The API and persistence boxes are a logical model inferred from observed requests and stored responses; server internals were not available.

Defect severity and priority

Supported defects by severity

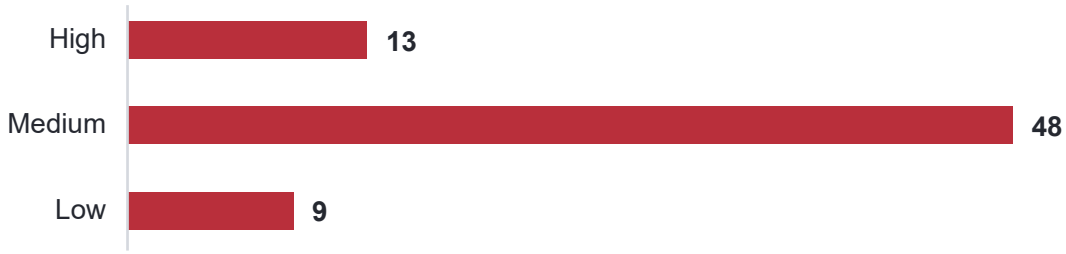


Figure 2 Counts use only the 70 supported defects. The 20 observations are excluded.

Severity	Count	Priority	Action
High	13	P1	Correct before enabling real customer ordering
Medium	48	P2	Repair in the next planned quality iteration
Low	9	P3	Correct copy and structure with focused regression

High-severity records

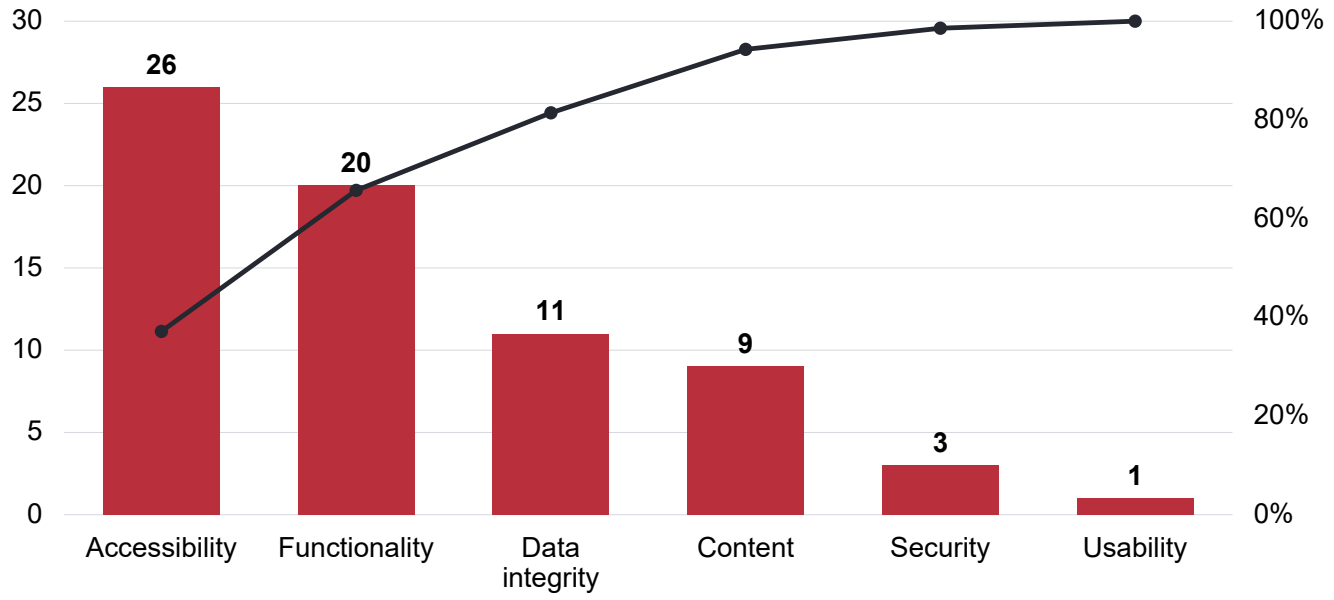
FM-021, FM-022, FM-024, FM-025, FM-028, FM-029, FM-030, FM-031, FM-032, FM-033, FM-034, FM-035, FM-040.

The severity describes observed impact; no Critical or Showstopper classification has been invented. Browser browsing continued, and monetary tests were limited to authorised demo COD orders.

Severity and priority are adapted from Lewis Chapter 24, pp. 265-266 (PDF pp. 292-293), and E12, p. 364 (PDF p. 391).

Pareto analysis of supported defects

Defects by category with cumulative percentage



Red bars: defect count Charcoal line: cumulative share of the 70 defects

Figure 3 Categories are inspection classifications, not statistically proven root causes.

Accessibility (26) and Functionality (20) together account for 46 of 70 defects (65.7%). This frequency helps organise improvement work, but the fewer High-severity price and data-integrity defects still require earlier action.

Category	Defects	Cumulative share
Accessibility	26	37.1%
Functionality	20	65.7%
Data integrity	11	81.4%
Content	9	94.3%
Security	3	98.6%
Usability	1	100.0%

Adapted from Lewis G21 Pareto Analysis and Exhibit G.43, pp. 499-501 (PDF pp. 526-528). The 80/20 relationship is not assumed to hold for this dataset.

Defect distribution by application area

Supported defects by module

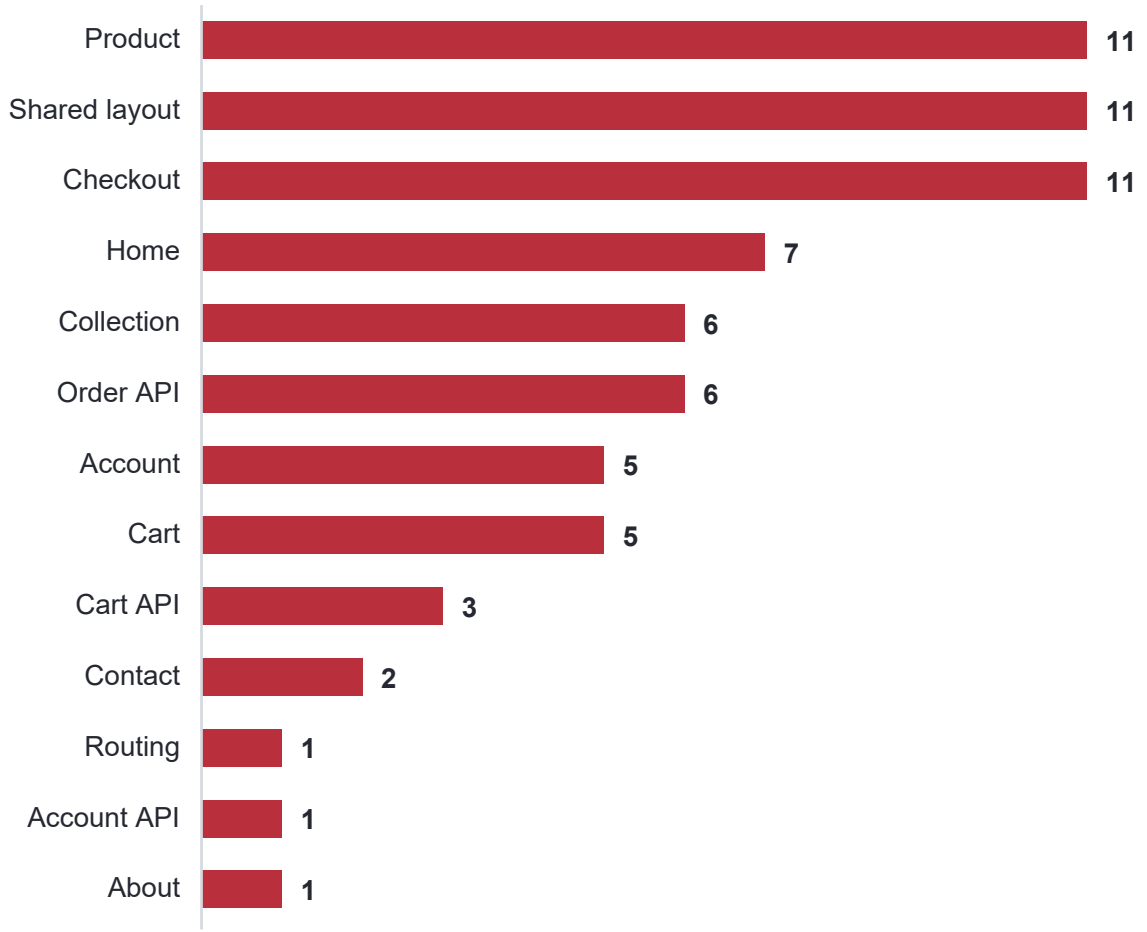


Figure 4 Each record is counted once in its primary module; shared symptoms are described in that record.

These counts identify where the audit encountered problems. They do not measure the relative quality of equally tested modules, because the amount of testing in each area was not equal.

A server validator change should trigger order and cart API retests. A shared navigation or title change should trigger all-route keyboard and responsive regression.

Adapted from the category and system-area analyses in Lewis Chapter 24, pp. 263-267 (PDF pp. 290-294), and E29, p. 385 (PDF p. 412).

Status and measurement boundaries

Current record status

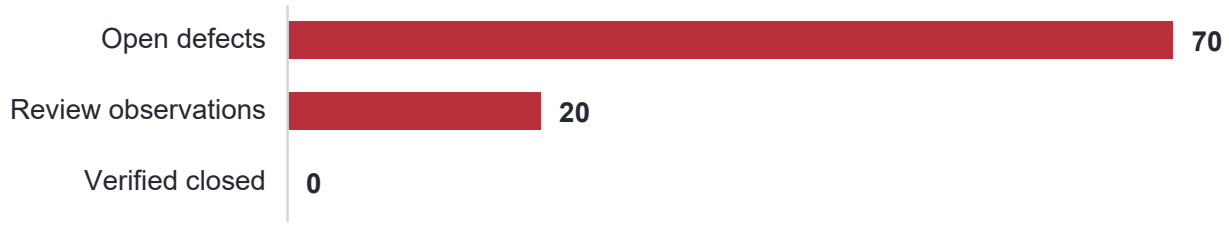


Figure 5 Status counts describe this audit handover, not a development-team dashboard.

Metric	Reported value or limitation
Requested defect count	100
Established distinct defects	70; target shortfall 30
Additional observations	20; not included in the defect count
Closure count	0; no repair or closure evidence received
Defect density per KSLOC	Not calculated: server source and valid source-size denominator unavailable
Test coverage percentage	Not calculated: no complete requirements or test denominator supplied
Defect age and closure trend	Not calculated: no repair history or closure dates
Performance and load curves	Not measured; no fabricated response-time, throughput or control-chart data

Lewis describes defect age and density on p. 267 (PDF p. 294) and tracking curves in Exhibit 24.1, p. 264 (PDF p. 291). Those charts require a time series or size data not present here; they have not been copied with invented values.

Cause and effect investigation

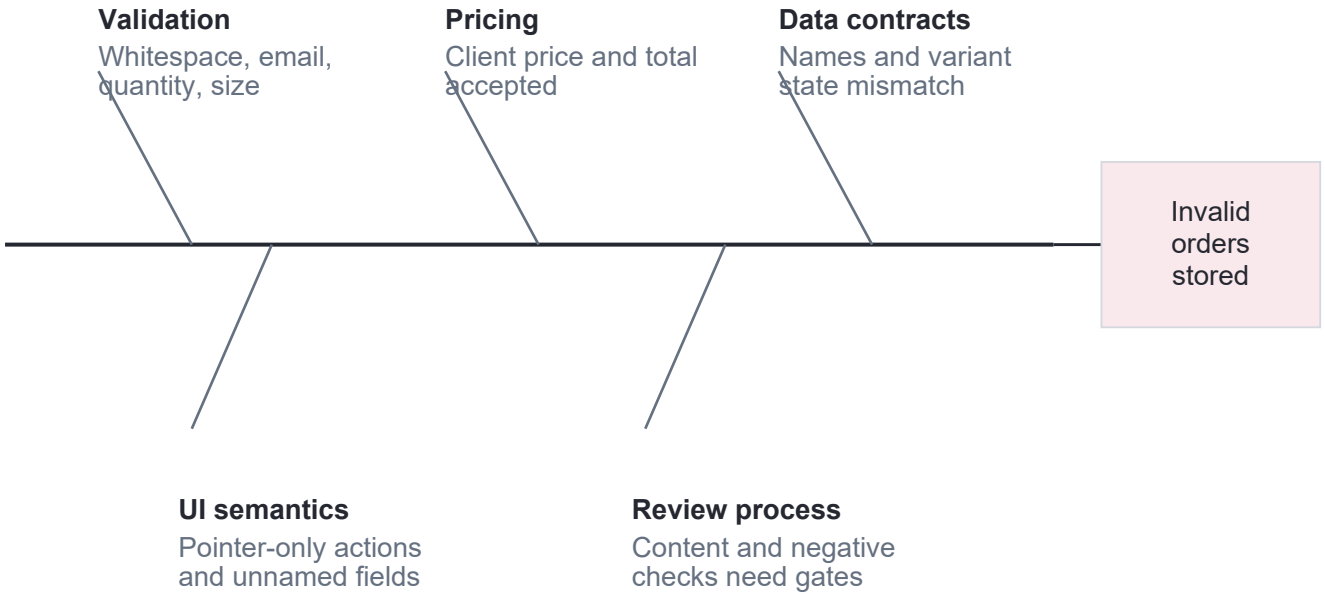


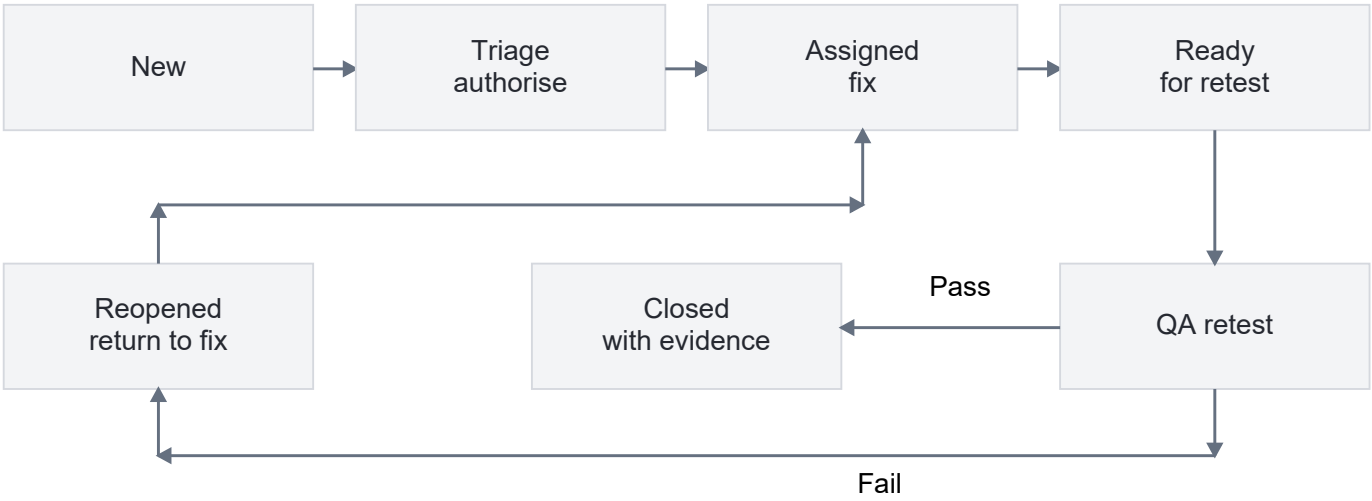
Figure 6 Investigation map adapted to ForeverMart. Branches are possible contributing factors; the server root cause has not been confirmed from source.

Evidence-backed factor	Investigation or prevention action
Order validation failures	Inspect request schemas and write-path checks for empty items, identity, variant, quantity and address
Monetary values persist as submitted	Locate server price calculation and determine why client money is accepted
Client data-contract mismatch	Standardise productName and variant state transitions across views
UI semantics and copied content	Review shared components with keyboard, labels and content checks before release

Confirm causes through server-code review and a reproduction after each fix. A common symptom or a large category count alone does not prove a single causal mechanism.

Adapted from Lewis cause-and-effect discussion, p. 42 (PDF p. 69), and G7 Cause-Effect Graphing, pp. 456-457 (PDF pp. 483-484).

Defect lifecycle and triage



Triage may mark Duplicate, Works as intended or New requirement with a reason.

Figure 7 Proposed defect handling workflow, adapted from Exhibit 24.2. It has not been executed by the development team.

Role	Proposed responsibility
Project owner	Confirm requirements, review observations and accept any residual risk
QA reviewer	Check evidence, reproduce, consolidate duplicates and specify retest acceptance
Developer	Investigate cause, implement the remedy and identify the fix build
QA retester	Retest the original scenario and related regression before closing

Closure evidence

A record closes only after the actual corrected build is identified, the original failure no longer occurs, relevant positive and negative checks pass, and fresh evidence is attached. If it fails, reopen it with the new actual result. No record in this report is marked fixed.

Adapted from Lewis Exhibit 24.2, p. 266 (PDF p. 293), defect meetings, p. 265 (PDF p. 292), and E12 status fields, p. 364 (PDF p. 391).

Continuous improvement plan

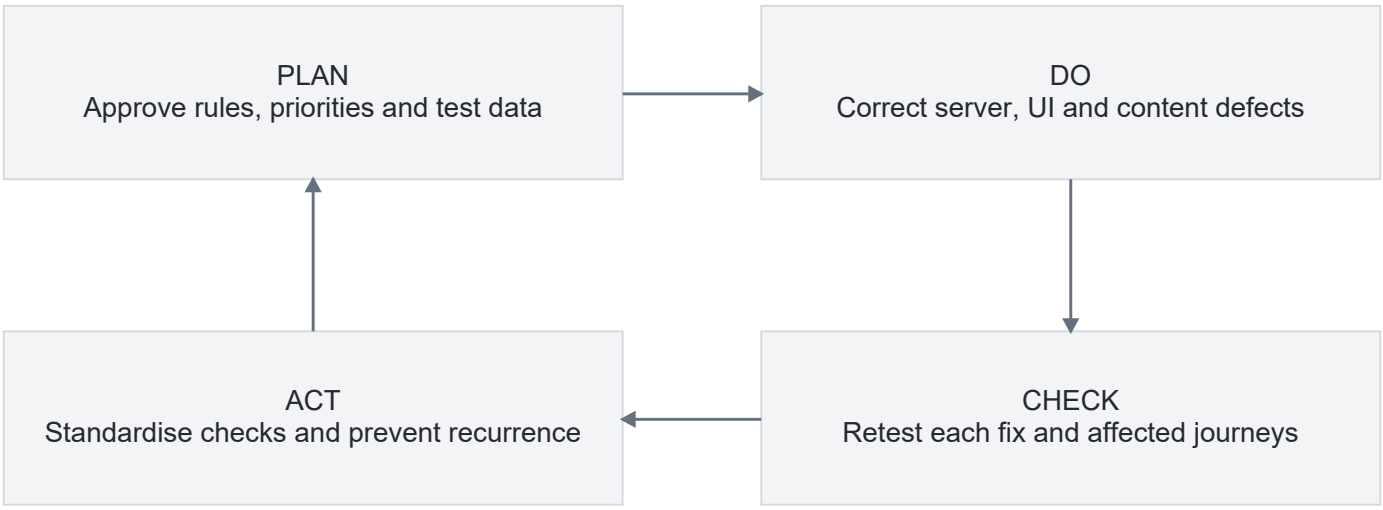


Figure 8 Project-specific Plan Do Check Act cycle. Proposed actions are not completed work.

Phase	Deliverable	Acceptance evidence
Plan	Approve domain rules, severity and supported countries/currency	Reviewed criteria and an owner decision for each observation
Do	Repair server validation/pricing, cart merge, controls and product copy	Identified fix build and developer change notes
Check	Run record-specific retests and critical shopping regression	Expected/actual results plus fresh screenshots or responses
Act	Add shared schemas, typed contracts, accessibility and content gates	Repeatable checks in the release workflow

Suggested work order: first protect server order/cart integrity and profile serialization; then restore missing actions and customer state; then correct shared accessibility and content; finally review the optional hardening and optimisation observations.

Adapted from Lewis Exhibit 3.1 and PDCA explanation, pp. 48-50 (PDF pp. 75-77), and E20 PDCA Test Schedule, p. 373 (PDF p. 400).

Corrective action and regression matrix

Changed component	Record references	Retest scope
Order validation and price calculation	FM-022-032	Valid COD order; invalid items and address; tampered money; confirmation totals
Cart validators and authentication merge	FM-020-021, 033-035	Guest and account carts; valid variants; bounds; unknown references; item display
Product and recommendation state	FM-004-011, 045, 055, 062	Accurate data; related navigation; gallery; variants; keyboard and alternatives
Collection/search state	FM-012-014, 041-042, 052, 056	Search normalization; empty state; filters/sort; Back; keyboard
Shared navigation and forms	FM-015-019, 043-044, 048-066	All routes; input names; focus; menu lifecycle; 320 px layout
Public content and configuration	FM-067-090 as applicable	Proofread pages; verify approved contacts/policies; check headers and metadata

Passing checks retained as regression baselines

The audit observed successful catalogue loading and normal image loading, correctly labelled category checkboxes, case-insensitive basic search, working price sorting, a size-required add-to-cart check, a quantity minimum of one in normal decrement controls, working signup/login/logout, 390 px reflow, a working favicon, HTTPS/HSTS and a 401 response when the own-cart request omitted authentication.

Approval remains pending

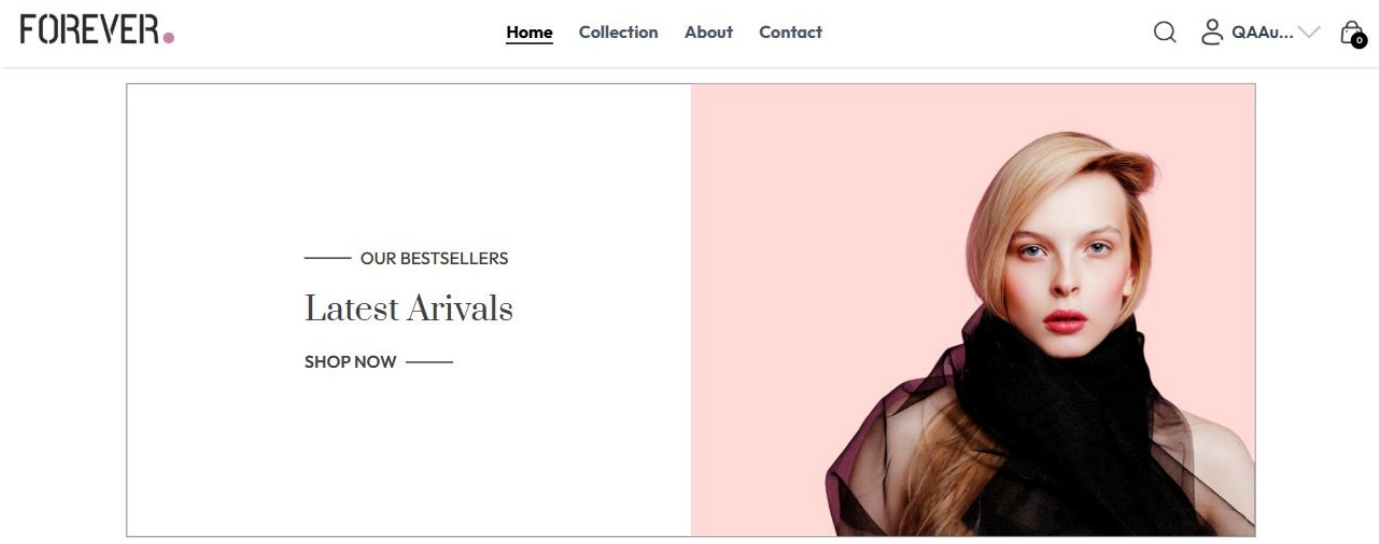
The project owner has not approved the inferred criteria, assigned remedies or accepted production risk. Developer and client comments remain Not supplied. Suggested delivery order is not a promised implementation schedule.

Adapted from Lewis E14 Retest Matrix, pp. 366-367 (PDF pp. 393-394). The matrix maps changes to tests; it does not claim those retests have been run.

Screenshot evidence | Hero action and spelling

FM-001 / FM-067

SHOP NOW did not navigate. The hero also misspells Arrivals.

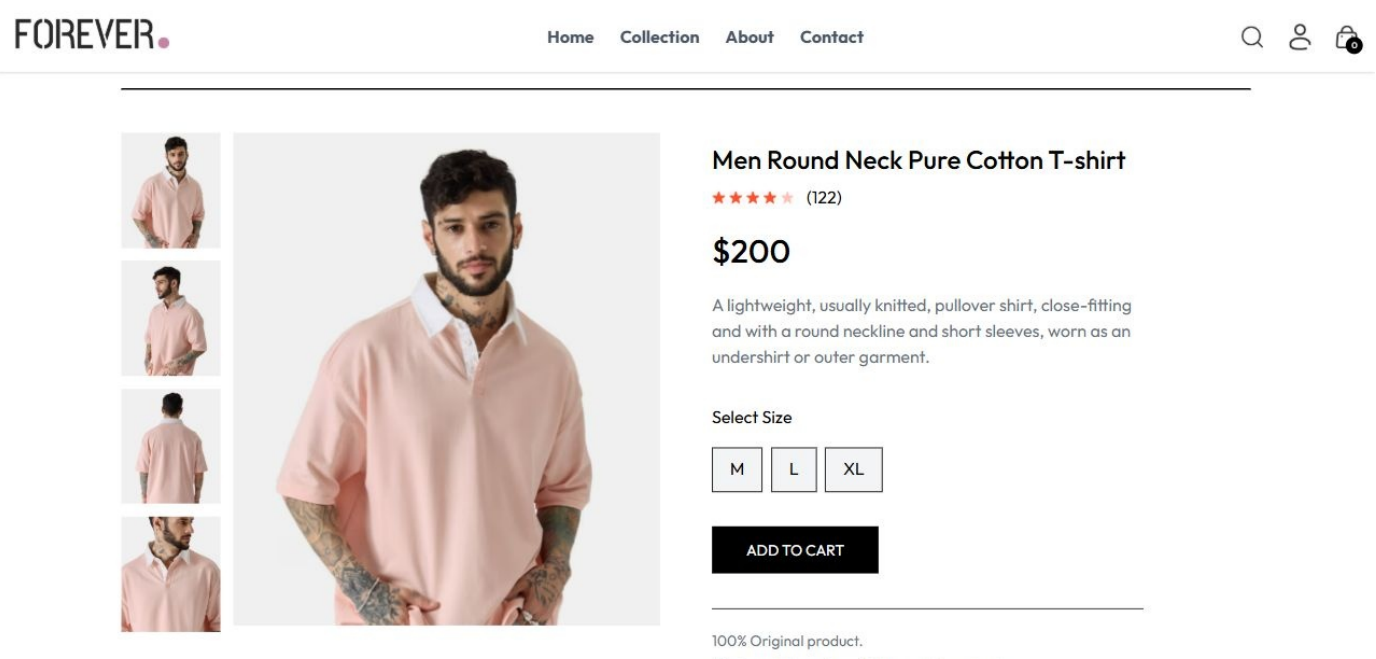


Excerpt from evidence/shop-now-no-action.jpg. The unmodified full capture is retained in the evidence folder.

Screenshot evidence | Product identity

FM-008

The inspected garment image and product identity do not agree.

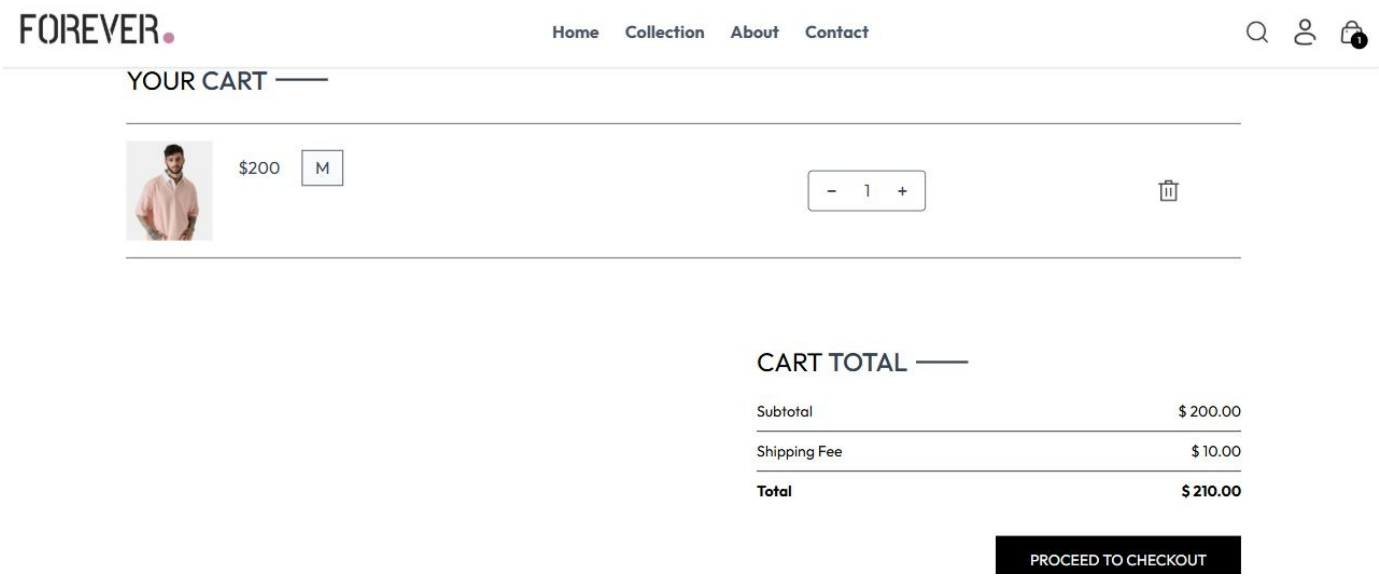


Excerpt from evidence/product-details.jpg. The unmodified full capture is retained in the evidence folder.

Screenshot evidence | Cart product names

FM-020

Cart row displays image and money while omitting the product name.

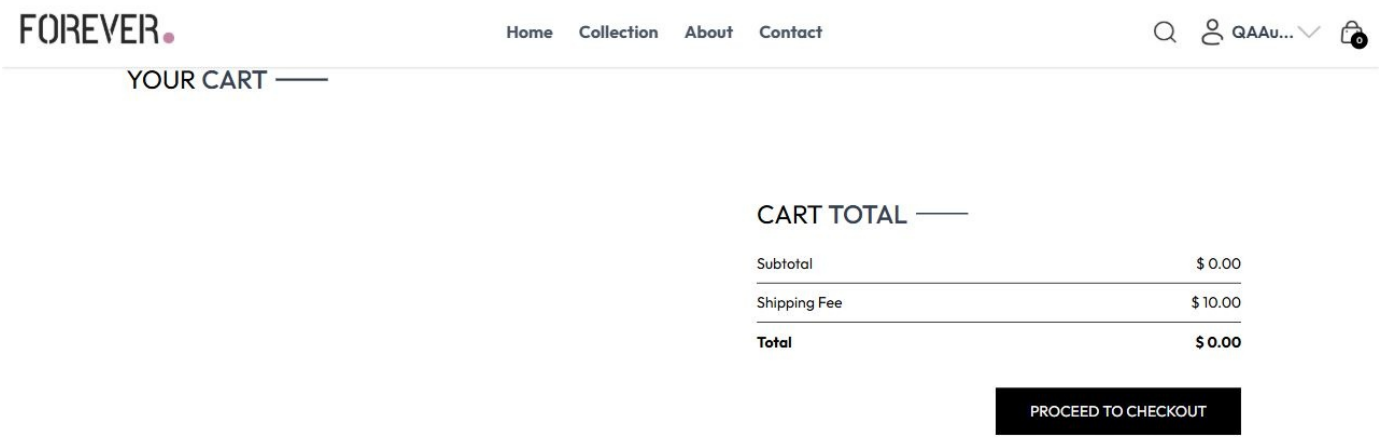


Excerpt from evidence/cart-one-item.jpg. The unmodified full capture is retained in the evidence folder.

Screenshot evidence | Empty-cart checkout

FM-022 / FM-023

An empty cart exposes checkout and inconsistent shipping/total presentation.

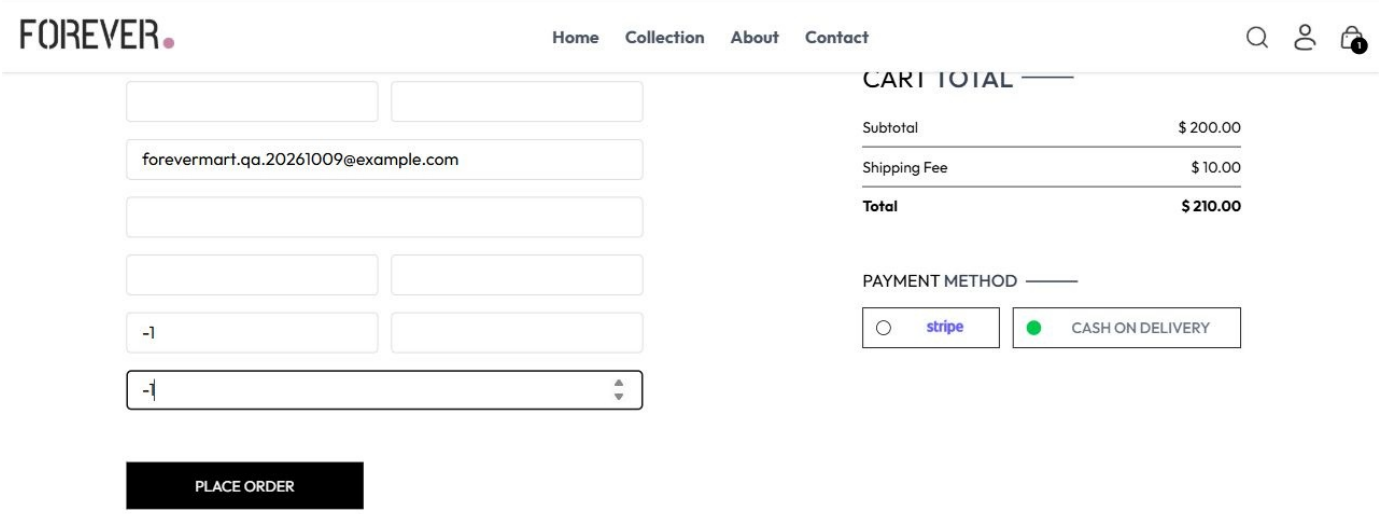


Excerpt from evidence/cart-empty.jpg. The unmodified full capture is retained in the evidence folder.

Screenshot evidence | Invalid delivery values

FM-024 / FM-025 / FM-026

The authorised demo COD test submitted whitespace strings and negative contact values.



Excerpt from evidence/checkout-invalid-values.jpg. The unmodified full capture is retained in the evidence folder.

Screenshot evidence | Incorrect jacket description

FM-006

The jacket page describes a short-sleeved shirt.

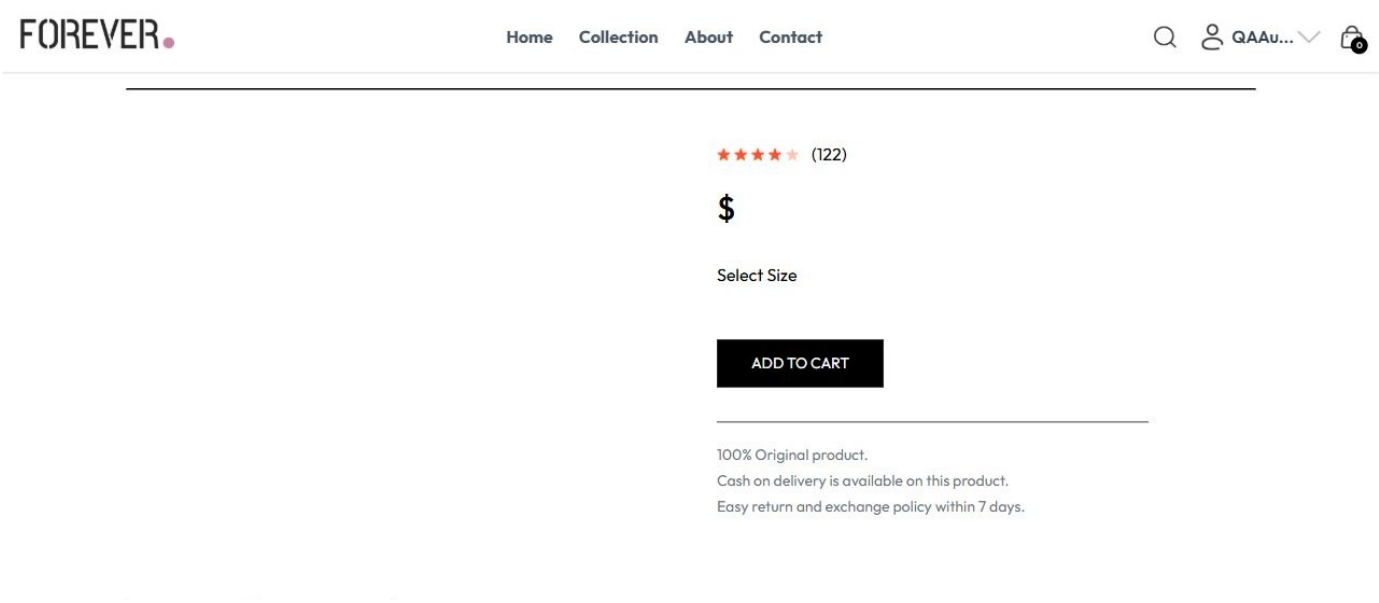


Excerpt from evidence/second-product-loaded.jpg. The unmodified full capture is retained in the evidence folder.

Screenshot evidence | Unknown product

FM-011

A missing product leaves an incomplete purchase page.

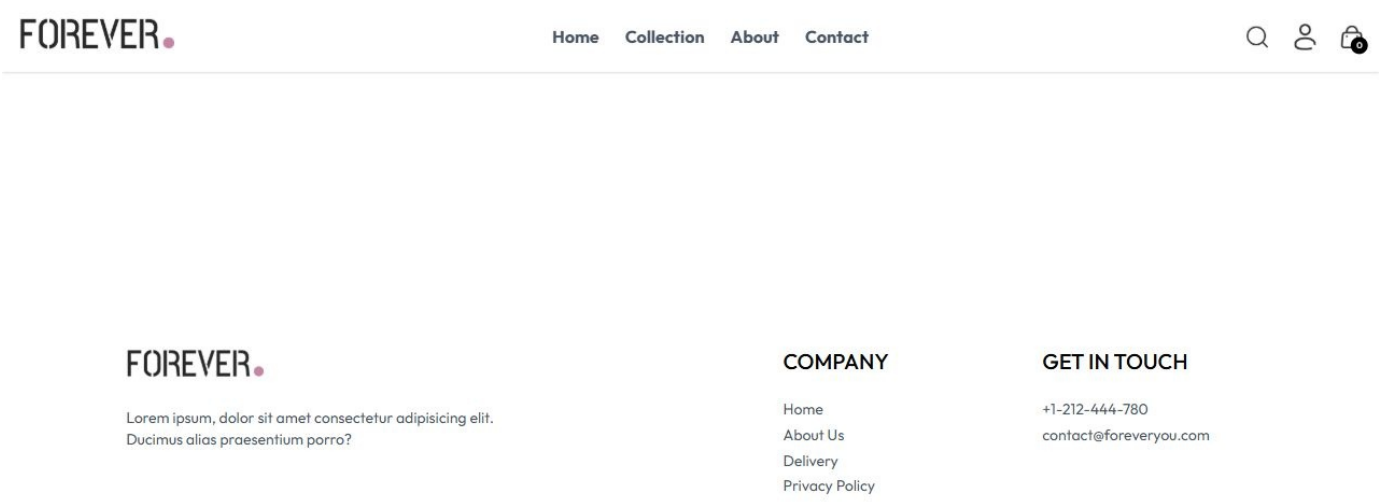


Excerpt from evidence/missing-product.jpg. The unmodified full capture is retained in the evidence folder.

Screenshot evidence | Unknown route

FM-039

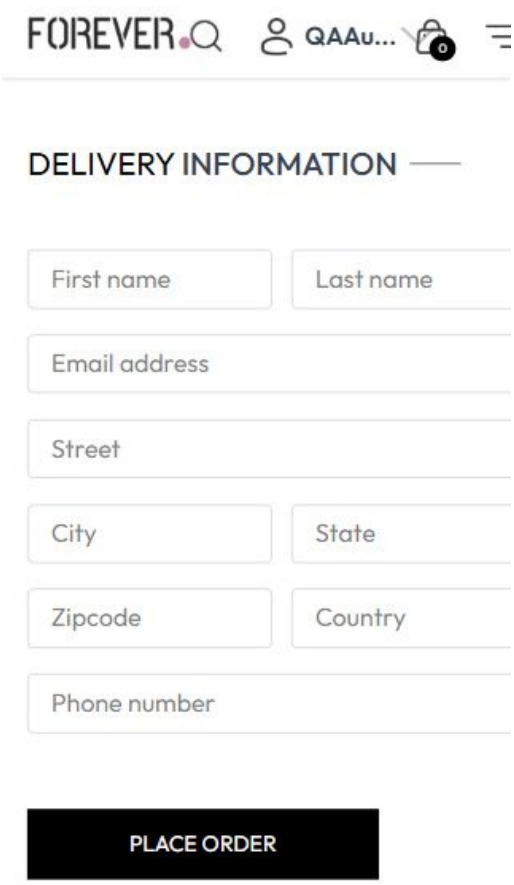
An unrecognised route offers no explanation or recovery body.

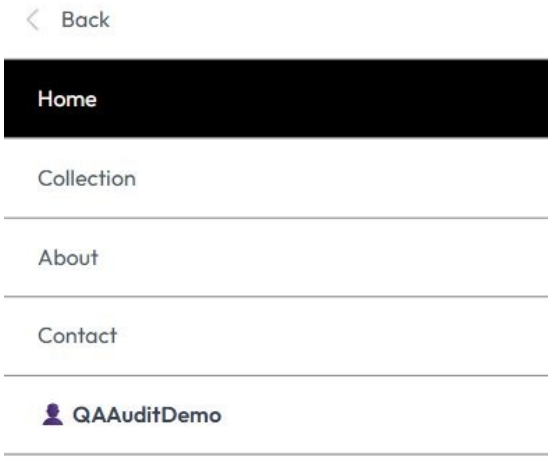


Excerpt from evidence/unknown-route.jpg. The unmodified full capture is retained in the evidence folder.

Screenshot evidence | Mobile and accessibility

FM-048 to FM-051 / FM-060 / FM-066





Measurement	Recorded result
320 px checkout	Document width 352 px: horizontal overflow
Muted home-page text	16 CSS px; contrast 2.60:1 on white (normal-text criterion 4.5:1)

Left: checkout-320-verified.jpg. Right: mobile-menu-open.jpg. Contrast uses recorded computed styles and a saved colour conversion.

API and persistence evidence

Test	Submitted boundary value	Response and read-back
Order total	Valid $200 item, quantity 1; amount=1	HTTP 200 success; amount 1 stored
Order line price	price=1, amount=11 for the $200 product	HTTP 200 success; altered price stored
Order quantity	quantity=-0.5	HTTP 200 success; quantity stored and displayed
Order variant	size=QA-NOT-A-SIZE	HTTP 200 success; unsupported size stored
Order identity	ID=000000000000000000000000	HTTP 200 success; fabricated reference stored
Order delivery email	address.email=not-an-email	HTTP 200 success; malformed value stored
UI delivery address	Whitespace-only required strings; phone=-1; zip=-1	Order stored with those values
UI empty order	items=[]	Order stored with amount 10
Cart boundary checks	Unknown ID, unsupported size and quantity=-0.5	Each accepted; invalid demo entries later removed
Authentication baseline	Own cart request with no authentication cookie	HTTP 401; no anonymous access demonstrated

Sensitive response fields

The own-profile response field names included password and refreshToken. Values were excluded from the evidence register and this document. Returning these fields is an exposure finding; this audit does not claim plaintext passwords or access to other users.

Evidence sources

raw/api-validation.json contains the six API order checks and own-order read-back. raw/demo-orders.json contains the earlier two UI orders. raw/cart-validation.json contains cart checks. All submitted identity and contact data are synthetic.

The server root cause is not asserted without source access. Remedies specify enforceable rules derived from the observed failures.

Reference book crosswalk

The attached reference is Software Testing and Continuous Quality Improvement, second edition, William E. Lewis, with Gunasekaran Veerapillai as technical contributor. Copyright 2005, Auerbach Publications / CRC Press, ISBN 0-8493-2524-2. Printed page n corresponds to PDF page n+27 for the cited numbered pages.

Book reference	Printed / PDF pages	Application in this report
E4 Internet/client-server test plan	353-354 / 380-381	Scope, assumptions, environment and exclusions
E6 test-case fields	358 / 385	Matching TC reference and expected/actual results
E11 system summary	361-362 / 388-389	Capabilities, deficiencies, risk and readiness opinion
E12 problem report	362-364 / 389-391	Detailed finding sheets and lifecycle metadata
E14 retest matrix	366-367 / 393-394	Change-to-regression mapping
E20 PDCA schedule; E21 strategy	373-375 / 400-402	Improvement stages and test approach
E27 defect table; E29 summary	383-385 / 410-412	Index, register, summary and classifications
Exhibit 24.2 defect lifecycle	266 / 293	Proposed triage, fix, retest, reopen and closure flow
Exhibit 3.1 PDCA circle	49 / 76	ForeverMart improvement cycle
G7 cause and effect	456-457 / 483-484	Hypothesis-based investigation diagram
G11 decision tables	492 / 519	Checkout decision table
G21; Exhibit G.43 Pareto	499-501 / 526-528	Actual category counts and cumulative percentage

Figures and tables are original adaptations populated with ForeverMart evidence. Unrelated book examples have not been copied. Run, scatter, control and defect-age charts require measurements not gathered in this audit, so no fabricated series is included.

The book was extracted in full for navigation, and the relevant reporting, management and testing-technique sections and representative layouts were reviewed.

Handover and supporting references

The course assignment requires testing a project, documenting defects and providing remedies. This report addresses Part 3 for ForeverMart. It does not replace the handwritten Part 1/2 work or claim the separate Python numerical simulation has been retested here.

Included deliverables

File or folder	Use
ForeverMart_Quality_Assurance_Report_Upgraded.pdf	Readable report with individual records, tables, figures and evidence excerpts
defect-registry.csv and .json	Editable complete register, suitable for filtering or importing into a defect tracker
ForeverMart_Quality_Assurance_Report.md	Editable text companion with the same detailed records
evidence and raw folders	Original captures and sanitized test evidence; filenames map from each finding
manifest.json; upgraded-page-map.json	Evidence hashes and original counts; updated report navigation map

Supporting standards and guidance

W3C WCAG 2.2 Labels or Instructions

W3C WCAG 2.2 Keyboard

W3C WCAG 2.2 Name Role Value

W3C WCAG 2.2 Contrast Minimum

W3C WCAG 2.2 Reflow

OWASP REST Security Cheat Sheet

MDN Content Security Policy

Next review

The owner should confirm the audit criteria, accept or reject the 20 observations, assign remedies, and provide a fix build for retesting. Further testing may establish additional defects; the requested total of 100 remains unfulfilled by 30 confirmed records.

Accessibility findings are targeted checks, not a complete WCAG conformance certification. HTTP-header observations are defence-in-depth recommendations, not demonstrated attacks.

Techniques, tools and reference sources

Type	Technique / tool / source	Use in this ForeverMart audit
Test method	Exploratory and state-transition testing	Browse, search, navigation, cart merge and related-product size state; reproduce visible failures.
Test method	Boundary values, equivalence partitions and negative tests	Empty values, whitespace, -1 contact fields, quantity -0.5, unknown IDs and unsupported sizes.
Test method	Authenticated API validation and read-back	Compare submitted data with saved orders/cart; expose altered prices, totals and invalid domains (FM-022 to FM-040).
Test method	Keyboard, DOM and responsive inspection	Check control semantics, labels, focus, alternative text and 320/390 px layouts (FM-041 to FM-066).
Test tool	Codex in-app browser and browser automation	Operate the demo site; capture screenshots, DOM snapshots, computed styles and authorised test responses.
Inspection	Public JavaScript, product JSON and HTTP headers	Corroborate UI/data contracts, catalogue content and deployment policy; saved as raw evidence.
Measurement	Python: JSON, math, counters and SHA-256	Calculate contrast, record totals and chart counts; validate evidence paths and file integrity.
Analysis	Decision table, traceability, Pareto and module/severity charts	Link test rules to findings and prioritise the 70 defects using observed counts.
Improvement	Cause/effect, defect lifecycle, PDCA and retest matrix	Structure proposed investigation, remedies, ownership and regression acceptance; repairs are not yet implemented.
Book	William E. Lewis, second edition, 2005	Software Testing and Continuous Quality Improvement. Ch. 24; E12/E14/E27/E29; G4/G11/G13/G21/G22. See the book crosswalk.
Standard	W3C WCAG 2.2	Keyboard, names/roles, labels, contrast and reflow criteria. Targeted checks rather than full certification.
Guidance	OWASP REST Security; MDN CSP	Server validation, response exposure and policy-hardening remedies. Links appear in supporting references.
Report tools	Python, ReportLab, pypdf, pdfplumber, PDFium and Pillow	Compose the document, preserve data, inspect text, render every page and verify layout. These are report tools, not defect scanners.